A Phase 2, Multicenter, Randomized, Controlled Clinical Trial Protocol

On-Premises LLM-Directed Robotic Pancreaticoduodenectomy with Perioperative Daraxonrasib (RMC-6236) in KRAS-Mutated Pancreatic Ductal Adenocarcinoma

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Keywords Pancreatic Ductal Adenocarcinoma; Robotic Whipple; Daraxonrasib; Physical AI; On-premises LLM; Randomized Controlled Trial; Multicenter; IND/IDE; Phase 2 Protocol; Co-Investment; VVUQ

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1 Statement of Compliance

This trial will be conducted in accordance with the protocol, the principles of the International Council for Harmonisation (ICH) Good Clinical Practice (GCP) guideline E6(R3), and all applicable United States regulations governing both the investigational drug and the investigational device, together with the Physical AI overlay that governs the on-premises large language model (LLM) directed robotic system. This Phase 2 study is the randomized, controlled, multicenter efficacy evaluation that the first-in-human combined Investigational New Drug (IND) and Investigational Device Exemption (IDE) protocol explicitly deferred [1]; the predicate study established the daraxonrasib recommended Phase 2 dose (RP2D) and device feasibility and safety, and this protocol now advances both components into a powered comparative trial. Two distinct but harmonized regulatory spines apply concurrently across the eight participating high-volume hepatopancreatobiliary (HPB) centers; both are carried in full below, and neither is subordinated to the other. The Sponsor-Investigator attests that the conduct of every study activity, the protection of every enrolled participant, and the integrity of every recorded datum will satisfy the more stringent of the two spines wherever they differ.

1.1 Drug Arm Compliance (IND, 21 CFR Part 312)

The perioperative daraxonrasib (RMC-6236) component is conducted under an IND in accordance with Title 21 of the Code of Federal Regulations. Specifically, the study complies with 21 CFR part 50 (Protection of Human Subjects, including the informed-consent requirements of §50.20 through §50.27) [2]; 21 CFR part 54 (Financial Disclosure by Clinical Investigators) [3]; 21 CFR part 56 (Institutional Review Boards); and 21 CFR part 312 (Investigational New Drug Application), including the sponsor and investigator responsibilities of subparts B and D, the safety-reporting requirements of §312.32, and the IND-application and amendment provisions of §312.20 through §312.31. Because the predicate Phase 1 study established the RP2D [1], the drug arm is no longer a 3+3 dose-finding evaluation; daraxonrasib proceeds at the fixed RP2D of 300 mg orally once daily as a Phase 2 controlled efficacy evaluation under §312.21(b). The IND remains in effect following the review period of §312.40, the protocol advance is filed as an amendment under §312.30, and no participant is dosed before the amended IND is in effect.

1.2 Device Arm Compliance (IDE, 21 CFR Part 812)

The on-premises LLM-directed eight-arm robotic pancreaticoduodenectomy (Whipple) system (PancreSpeed II) is an investigational medical device evaluated under an IDE in accordance with 21 CFR part 812 [4]. The Sponsor-Investigator and the reviewing Institutional Review Board (IRB) have determined that the device presents a significant risk within the meaning of §812.3(m), because it is a surgical-intervention device whose patient-contact motion, force application, vascular-exclusion gating, and emergency-stop behavior present a potential for serious risk to the health, safety, or welfare of a participant. The study therefore proceeds only under an IDE approved by both the FDA and the IRB. The device arm additionally complies with 21 CFR part 50 [2] and 21 CFR part 56 for human-subject protection and IRB oversight, and with 21 CFR part 11 (Electronic Records; Electronic Signatures) [5] for the validation, audit-trail, and electronic-signature controls that underpin the federated hash-chained command and telemetry record described in §10. Having met the early-feasibility objectives of the predicate study [1], the device component is now framed as a multicenter randomized controlled clinical study under part 812, in which the comparative clinical performance of the device-directed operation is evaluated against institutional-standard high-volume pancreaticoduodenectomy.

1.3 Physical AI Overlay (21 CFR Part 312, Subpart J)

The autonomy-bearing behavior of the LLM-directed system is governed by the Physical AI overlay codified in 21 CFR part 312, Subpart J (§312.400 through §312.405), as adapted for end-to-end Physical AI oncology investigations [6]. For this Phase 2 study the Phase 0 gate is upgraded relative to the predicate. As preconditions to randomized enrollment, Subpart J now requires at least 5000 simulated procedures reproduced across at least three independent simulation frameworks, with a tightened sim-to-real correspondence of trajectory error < 1.5 mm and tip-force error < 0.4 N; a documented Unified Safety Level (USL) readiness rating of ≥ 8.0 for surgical deployment, raised from the predicate threshold; per-site qualification together with multicenter fleet harmonization so that every device instance behaves identically across the eight centers; and a federated hash-chained audit trail spanning all sites under 21 CFR part 11 [5]. The system remains classified as a Class II collaborative device under continuous human oversight with a guaranteed emergency-stop latency of ≤ 500 ms, operated as Stage 2 supervised semiautonomy with immediate manual override; full autonomy is prohibited under 21 CFR §312.21(e). These overlay requirements operate in addition to, and never in place of, the part 312 and part 812 obligations above. The hash-chained audit trail required under §312.404 satisfies the part 11 electronic-records standard and renders every LLM advisory, every robot command, and every safety-limit event re-traceable to a deterministic seed and a specific repository commit.

1.4 Good Clinical Practice and Electronic Records

The trial adheres to ICH E6(R3) GCP, the international ethical and scientific quality standard for designing, conducting, recording, and reporting clinical investigations involving human participants. All electronic clinical data, source records, and the cyber-physical telemetry stream are maintained under 21 CFR part 11 [5], with validated systems, secure time-stamped audit trails that do not obscure prior entries, role-based access controls, and electronic signatures bound to the records they authenticate. The federated hash-chained record specified by the Physical AI overlay provides cryptographic continuity across the screening, operative, and follow-up phases and across all eight participating sites, so that the multicenter telemetry corpus forms a single tamper-evident chain of custody.

1.5 Multicenter Single IRB and Co-Investment Governance

Because this is a cooperative multicenter study, a single IRB of record (sIRB) reviews the protocol for all eight participating sites in accordance with 45 CFR 46.114 [7], with site-level context handled through reliance agreements coordinated by the Coordinating Center. All co-investment in the trial is disclosed and managed under 21 CFR part 54 [3]. Contributions to the Patient-Aligned Co-Investment Facility (PACIF) are held behind a capital firewall that isolates every funder from randomization, endpoint definition, outcome adjudication, analysis, and publication; capital may purchase only operational capacity and never scientific outcomes. This separation is governed under the H.R. 9510 verification, validation, and uncertainty quantification (VVUQ) financial-data standard [8], which subjects the flow and application of co-investment funds to the same auditable VVUQ discipline applied to the trial's technical evidence.

1.6 Trial Registration

The study will be registered on ClinicalTrials.gov before enrollment of the first participant. It is registered as an Interventional study with a randomized, parallel-assignment allocation. The drug component is classified as Phase 2 (controlled efficacy evaluation of daraxonrasib at the established RP2D), while the device component carries Phase: Not Applicable, consistent with the ClinicalTrials.gov convention that FDA-defined phase categories apply to drugs and biologics rather than to devices. The Primary Purpose is recorded as Treatment. The study is open-label with blinded independent central review (BICR) of imaging, central masked pathology, and blinded endpoint adjudication, and it carries prespecified group-sequential interim analysis with efficacy and futility stopping rules.

1.7 Statement of Compliance Pathway

Figure 1 renders the combined IND and IDE compliance spine with the Physical AI Subpart J overlay as upgraded for Phase 2. The daraxonrasib drug arm (IND, part 312, now at the established RP2D) and the robotic Whipple device arm (IDE, part 812, significant-risk per §812.3(m), now a randomized controlled study) converge through the upgraded Phase 0 simulation-validation gate, the raised USL readiness verification, the single IRB across the eight sites, and the Class II collaborative classification into a single multicenter randomized enrollment pathway.

1.8 Signatures

The Sponsor-Investigator and the Coordinating Principal Investigator have reviewed and approved this protocol and confirm that the study will be conducted in compliance with the protocol, ICH E6(R3) GCP, and the federal regulations cited above. By signature below, each signatory agrees to conduct or supervise the study in accordance with the protocol and all applicable requirements.

Sponsor-Investigator	**Coordinating Principal Investigator**
______________________	______________________
Kevin Kawchak	Name (printed)
CEO, ChemicalQDevice	Title / Institution
Signature and date: ____________	Signature and date: ____________

Multi-Site Principal Investigator	**Site / Institution**
______________________	______________________
Name (printed)	One signature line per participating site
Title / Institution	(eight high-volume HPB centers)
Signature and date: ____________	Signature and date: ____________

The signatories acknowledge that this is an independent research draft; it is not an active IND or IDE and is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor.

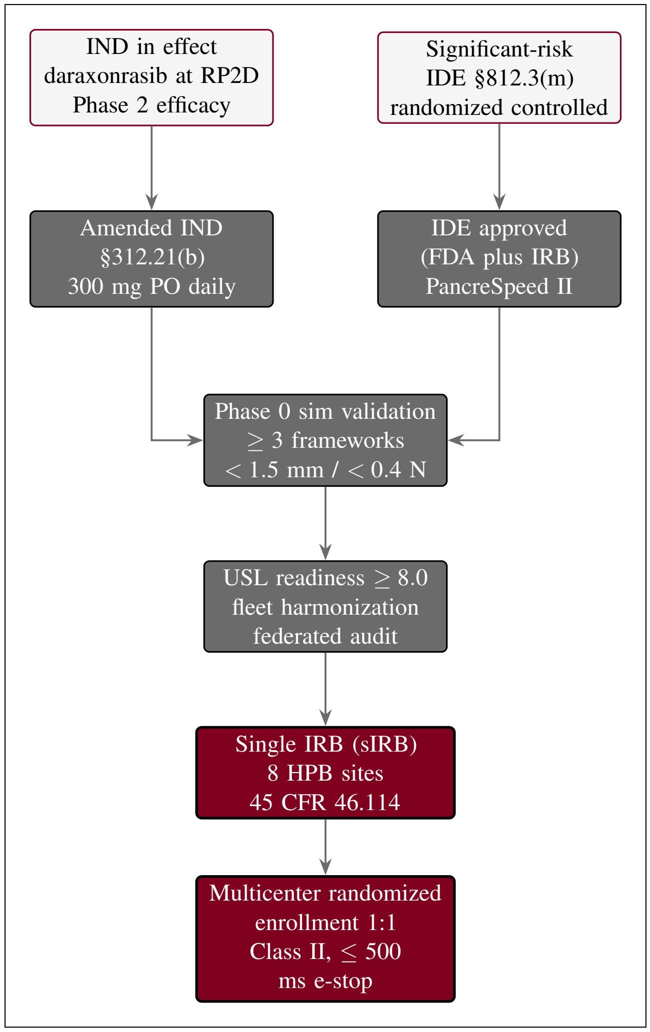

Figure 1. Combined IND / IDE regulatory pathway with the upgraded Physical AI overlay for Phase 2. The daraxonrasib drug arm (IND, part 312, at the established RP2D) and the robotic Whipple device arm (IDE, part 812, significant-risk per §812.3(m), randomized controlled) converge through the upgraded Subpart J overlay of Phase 0 simulation validation across at least three frameworks, USL readiness ≥ 8.0 with multicenter fleet harmonization, single IRB review across eight sites, and Class II collaborative classification into a single multicenter randomized enrollment pathway.

2 Protocol Summary

2.1 Synopsis

Title. A Phase 2, multicenter, randomized, controlled clinical trial of on-premises large language model (LLM) directed robotic pancreaticoduodenectomy (Whipple) with perioperative daraxonrasib (RMC-6236) in KRAS-mutated pancreatic ductal adenocarcinoma (PDAC).

Rationale. PDAC is the third leading cause of cancer death in the United States, with total five-year overall survival below 13 percent [9], and even at high-volume centers a substantial fraction of intended curative resections fail to achieve a margin-negative result or are compromised by perioperative morbidity [10]. The first-in-human predicate study established the daraxonrasib RP2D and demonstrated the feasibility and hard cyber-physical safety of an on-premises LLM-directed robotic Whipple [1], but it was not powered for efficacy and explicitly deferred the comparative question to a randomized trial. This Phase 2 study pairs perioperative daraxonrasib at the established RP2D with the upgraded PancreSpeed II device-directed operation and tests, against contemporary standard of care, whether the combined intervention improves disease control after curative-intent surgery.

Objectives and endpoints. The primary objective is to determine whether the experimental intervention prolongs progression-free survival (PFS) relative to standard of care. The primary endpoint is PFS per RECIST 1.1, investigator assessed with blinded independent central review (BICR) as a sensitivity analysis. Key secondary endpoints, tested in a prespecified hierarchical sequence, are overall survival (OS); the R0 resection rate; the International Study Group on Pancreatic Surgery (ISGPS) grade B/C postoperative pancreatic fistula rate [11]; major pathologic response (MPR); and KRAS circulating tumor DNA (ctDNA) clearance at week 12 [12].

Design. Multicenter, randomized 1:1, parallel-group, controlled, open-label study with BICR of imaging, central masked pathology, and blinded endpoint adjudication, conducted at eight high-volume HPB centers with a planned sample size of $n = 220$ randomized participants (approximately 110 per arm). Central, permuted-block randomization is stratified by resectability (resectable versus borderline-resectable), KRAS allele (G12D versus other G12), neoadjuvant therapy (yes/no), and site. The trial targets approximately 140 PFS events for 85 percent power at a two-sided alpha of 0.05 to detect a hazard ratio of 0.60 (median PFS 14.0 months control versus 23.3 months experimental), with a single interim analysis for efficacy and futility at approximately 60 percent of events by an independent Data and Safety Monitoring Board using O'Brien-Fleming spending [13]. A mandatory upgraded Phase 0 simulation-validation gate (≥ 5000 simulated procedures across ≥ 3 independent frameworks, sim-to-real trajectory < 1.5 mm and tip-force < 0.4 N, USL ≥ 8.0, and multicenter fleet harmonization) precedes any device-directed activity in the experimental arm.

Population. Adults ($\geq$ 18 years) with histologically confirmed, resectable or borderline-resectable, KRAS G12-mutated PDAC; ECOG performance status 0 or 1; and adequate organ function. Participants are daraxonrasib-eligible under the broad pan-KRAS criteria of the RASolute 302 program [14]. Full inclusion and exclusion criteria appear in §5.

Intervention. Participants are randomized 1:1 to one of two arms. Arm A (experimental) receives perioperative daraxonrasib at the RP2D of 300 mg orally once daily plus an on-premises LLM-directed eight-arm robotic Whipple (PancreSpeed II) performed under continuous human oversight as a Class II collaborative device with hard cyber-physical limits. Arm B (control) receives standard perioperative systemic therapy (modified FOLFIRINOX [15]) plus an institutional-standard high-volume pancreaticoduodenectomy that is not LLM-directed.

Co-investment. The Patient-Aligned Co-Investment Facility, operating behind a capital firewall that bars any funder from influencing randomization, endpoints, adjudication, analysis, or publication, funds only the operational levers that make the eight-site, adequately powered, and equitable design possible.

Duration. Each participant undergoes up to 28 days of screening, central randomization with a baseline assessment, surgery on day 0, an acute day 1 to 7 window, the day-30 safety window, a day-90 pathology and MPR assessment, and long-term follow-up every 8 to 12 weeks with BICR imaging to the PFS endpoint and the 24-month overall-survival endpoint.

2.2 Schema

Figure 2 presents the end-to-end randomized multicenter schema, from 28-day screening through eligibility determination, central 1:1 stratified randomization, the two parallel treatment arms (Arm A daraxonrasib at RP2D plus LLM-directed robotic Whipple; Arm B modified FOLFIRINOX plus standard Whipple), follow-up with BICR, and the PFS and overall-survival endpoints, including the screen-failure branch.

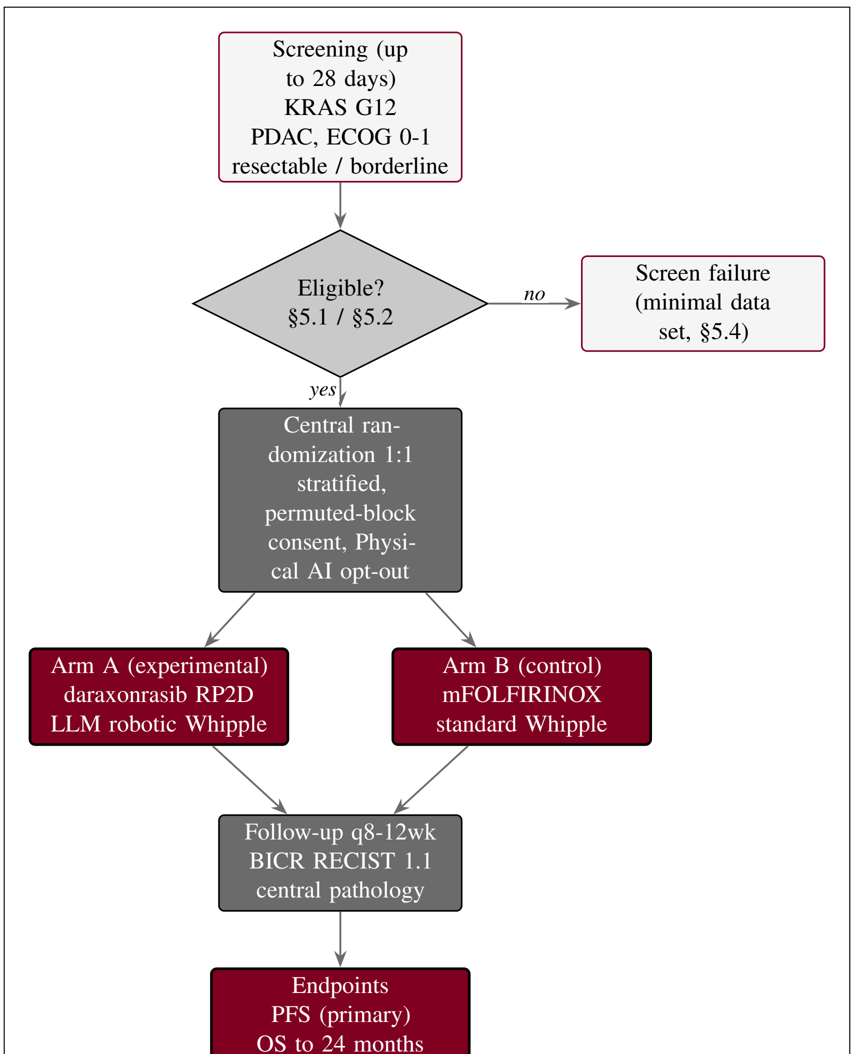

Figure 2. Overall schema for the Phase 2 multicenter randomized study, from 28-day screening through eligibility determination, central 1:1 stratified randomization with the Physical AI consent opt-out, the two parallel arms (Arm A daraxonrasib at RP2D plus LLM-directed robotic Whipple; Arm B modified FOLFIRINOX plus standard Whipple), follow-up with blinded independent central review, and the progression-free survival and 24-month overall-survival endpoints, including the screen-failure branch.

2.3 Schedule of Activities (SoA)

The Schedule of Activities binds each assessment to a timepoint across the screening window (Day -28 to -1), randomization and baseline, surgery (Day 0), the acute window (Day 1 to 7), Day 30, Day 90, and long-term follow-up every 8 to 12 weeks. An “X” marks each assessment performed at that visit. Arm-specific rows are flagged in the activity label: the Phase 0 simulation sign-off with USL $\geq$ 8.0 verification, the intra-operative telemetry capture, and the daraxonrasib restart advisory apply to Arm A, while ctDNA (KRAS) is drawn at baseline and week 12 in both arms. The full-width Schedule of Activities is given in Table 1.

Table 1: Schedule of Activities for the Phase 2 multicenter randomized trial.

Activity	Screen Day -28 to -1	Rand / Base	Surg Day 0	Acute Day 1-7	Day 30	Day 90	LT q8-12wk
Informed consent (incl. Physical AI opt-out)	X						
Staging / KRAS G12 confirmation	X						
CA 19-9	X	X			X	X	X
ECOG performance status	X	X			X	X	X
Safety laboratory panel	X	X	X	X	X	X	X
ctDNA (KRAS)		X				X	
Central randomization (1:1, stratified)		X					
Phase 0 sign-off, USL $\geq$ 8.0 (Arm A)		X					
Robotic / standard Whipple surgery			X				
Intra-op telemetry capture (Arm A)			X				
ISGPS fistula grading				X	X		
Central pathology R0 / MPR						X	
Daraxonrasib restart advisory (Arm A)				X			
QoL EORTC QLQ-C30		X			X	X	X
BICR RECIST imaging	X					X	X
AE / SAE review	X	X	X	X	X	X	X
PFS / OS assessment						X	X

Abbreviations: AE, adverse event; SAE, serious adverse event; BICR, blinded independent central review; ctDNA, circulating tumor DNA; EORTC QLQ-C30, European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire Core 30 [16]; ISGPS, International Study Group on Pancreatic Surgery; LT, long-term; MPR, major pathologic response; OS, overall survival; PFS, progression-free survival; QoL, quality of life; RECIST, Response Evaluation Criteria in Solid Tumors; USL, Unified Safety Level. Randomization and baseline assessments occur on the same visit before surgery; Day 30 is the primary safety window; Day 90 carries central R0 and MPR pathology; arm-flagged rows apply to the experimental arm (Arm A).

3 Introduction

3.1 Study Rationale

Pancreatic ductal adenocarcinoma (PDAC) is the third leading cause of cancer death in the United States and the fourth in the European Union, with total five-year overall survival below 13 percent [9]. The pancreaticoduodenectomy, or Whipple procedure, is the most technically demanding of all curative-intent abdominal oncology operations: it requires resection and reconstruction of four luminal structures (the pancreas, the duodenum, the common bile duct, and the stomach or pylorus) and dissection around two named vessels (the superior mesenteric vein and the portal vein), with an artery-first approach to the superior mesenteric artery. The leading lethal postoperative sequence is the fistula-sepsis cascade: pancreatic fistula, intra-abdominal infection, hemorrhage, sepsis, and failure to rescue, graded under the ISGPS classification of postoperative pancreatic fistula [11].

The 2025 Dutch nationwide cohort of 1000 robotic pancreaticoduodenectomies reported a conversion rate of 10.1 percent, Clavien-Dindo grade III or higher complications of 41.3 percent, ISGPS grade B or C fistula of 24.4 percent, and 30-day mortality of 3.9 percent [10]; these figures anchor the contemporary human baseline. The Phase 1 predicate [1] established the recommended Phase 2 dose (RP2D, 300 mg daraxonrasib once daily) and the feasibility and safety of the on-premises large language model (LLM) directed eight-arm robotic Whipple, but a single-arm feasibility study cannot establish efficacy. That predicate result creates genuine clinical equipoise between the integrated combination and the prevailing standard of care, and it is precisely that equipoise that this randomized, controlled efficacy comparison is designed to resolve. The present Phase 2 protocol therefore randomizes patients 1:1 (open-label, with blinded independent central review, BICR) between Arm A, perioperative daraxonrasib at RP2D plus the on-premises LLM-directed robotic Whipple (PancreSpeed II), and Arm B, modified FOLFIRINOX [15] plus a standard high-volume pancreaticoduodenectomy.

The clinical argument for the combination, and for funding a definitive test of it, is best stated through four counterfactual scenarios, summarized in Figure 3, in which withholding the integrated approach shortens progression-free survival (PFS) and overall survival (OS) or, in the fourth case, denies the field a trustworthy answer altogether. In Scenario A, a borderline-resectable tumor abutting the superior mesenteric vein progresses to unresectable disease during a human-only scheduling delay (operating room access, fatigue, staging lag), and the R0 window closes; the LLM-directed robot instead achieves a rapid, precise, in-window R0 resection. In Scenario B, an unrecognized superior-mesenteric or portal-vein injury during manual surgery triggers the hemorrhage-conversion-fistula-sepsis-failure-to-rescue cascade; the five-vessel no-fly vascular gate with a ≤ 3 ms cross-arm emergency stop averts the injury and preserves a complete resection. In Scenario C, manual mistiming of the daraxonrasib restart (too early or too late) causes anastomotic dehiscence or micrometastatic outgrowth; the LLM advisory restart at T+7, T+14, or T+21 days, keyed to fistula status and drug trough, preserves the therapeutic window. In Scenario D, an under-funded, single-center, under-powered study fails to deliver a definitive, generalizable, and equitable answer at all: slow accrual, attrition, and a thin biomarker core leave the central PFS question unresolved for the next patient. The patient-aligned co-investment model converts capital, strictly behind a structural firewall, into accrual speed, retention, fidelity, and statistical power, raising the probability of a definitive result without ever touching the endpoints, their adjudication, or the analysis.

These scenarios invite four framing questions that this protocol is designed to answer constructively rather than rhetorically. First, given the narrow resection window and the unforgiving fistula-sepsis cascade, how did we ever rely on humans alone for this treatment, with no second-opinion oracle isolated from the operative kinematics. Second, why did we fully trust a prior trial system with so many opportunities for compounding human error across scheduling, dissection, and drug timing, when each step admits a documented, deterministic safeguard. Third, as AI outputs grow more complex and voluminous, why would human peer review alone continue to suffice for increasingly intricate AI-directed care, rather than a verification-before-generation discipline with a hash-chained audit trail [8]. Fourth, when aligned philanthropic and private capital can raise the success likelihood of a definitive trial without altering a single endpoint, why would we leave that answer under-powered and inequitable. The protocol answers the first three by retaining the human in the loop while making every machine action validated, bounded, and re-traceable, and it answers the fourth with a Patient-Aligned Co-Investment Facility whose capital firewall lets contributions buy only operational success-likelihood, never a favorable result [8].

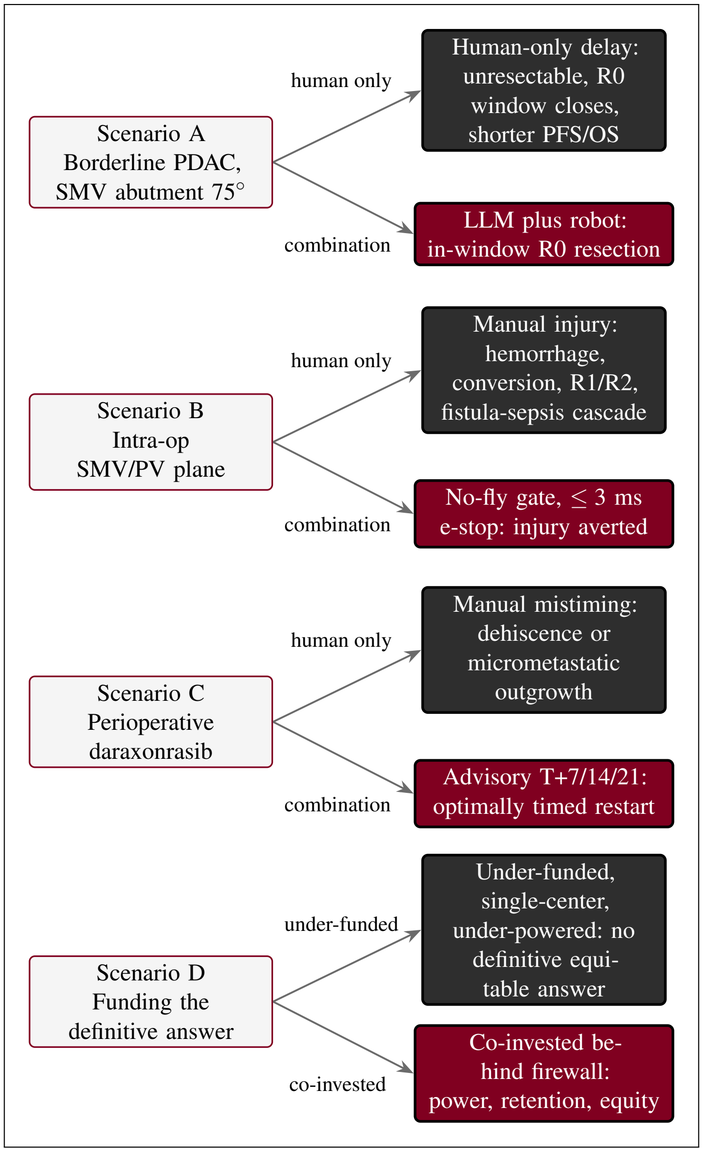

Figure 3. The four counterfactual scenarios in which withholding the integrated approach worsens the patient or the evidence. Scenario A: a human-only scheduling delay lets a borderline-resectable tumor progress and closes the R0 window. Scenario B: an unrecognized superior-mesenteric or portal-vein injury during manual surgery triggers the fistula-sepsis cascade; the no-fly gate with a ≤ 3 ms emergency stop averts it. Scenario C: manual mistiming of the daraxonrasib restart causes dehiscence or micrometastatic outgrowth; the timed advisory prevents it. Scenario D: an under-funded, single-center, under-powered study yields no definitive equitable answer; patient-aligned co-investment, walled off from the endpoints by the capital firewall, converts capital into accrual speed, retention, fidelity, and statistical power.

3.2 Background

PDAC epidemiology and the Whipple fistula-sepsis cascade

The epidemiologic burden and operative hazard of PDAC define the unmet need this protocol addresses. With under-13-percent five-year survival [9] and a most-demanding curative operation whose leading complication is a self-amplifying fistula-sepsis-hemorrhage cascade [11], the population has few alternatives and a narrow margin for technical error. The contemporary robotic baseline [10] demonstrates that even expert, high-volume robotic surgery leaves a substantial residual of conversions, high-grade complications, and clinically relevant fistulae. The combination in this protocol targets precisely the failure modes that the baseline still tolerates, and the randomized internal control (Arm B) measures the increment against modified FOLFIRINOX plus standard pancreaticoduodenectomy [15].

Daraxonrasib mechanism and clinical program

Daraxonrasib (RMC-6236) is an oral RAS(ON) multi-selective inhibitor that binds the active, GTP-bound (ON) state of multiple RAS variants, providing broad pan-KRAS coverage including the G12 alleles that predominate in PDAC. The FDA granted daraxonrasib Breakthrough Therapy designation in June 2025 for previously treated metastatic PDAC with KRAS G12 mutations [17], and the late-phase program comprises RASolute 302 (a Phase 3 study in previously treated metastatic PDAC) and RASolve 301 (a first-line program) [14]. The broad pan-KRAS eligibility of RASolute 302 defines the molecular eligibility used in this protocol, and the perioperative pause-and-restart logic respects any RASolve 301 checkpoint-inhibitor combination context. Five prior computational works by the author independently complemented this drug-development program throughout 2025 and 2026 and supply the simulation scaffolding adapted here: a 60-second PDAC robotic Whipple and daraxonrasib simulation [18]; an AI digital-twin pancreatic-cancer simulation for in silico FDA-compliance and cost efficiency [19]; a quantitative systems pharmacology (QSP) metastatic-PDAC trial simulation with code, VVUQ, and playbook [20]; a 100,000-patient in silico Phase 3 five-arm PDAC trial triplicate [21]; and end-to-end PDAC digital-twin clinical-trial proposals [22].

Current state of AI in clinical trials and surgery

Artificial intelligence in regulated medicine has, to date, been narrow and prediction-oriented rather than generative and intervention-directing. As of the current taxonomy of FDA-authorized medical devices, AI is overwhelmingly deployed as prediction-only or narrow task-specific software, with reported authorizations spanning 1,016 in the published taxonomy [23] and rising to 1,450 or more by current counts, while zero generative LLM devices have been approved for patient-facing intervention. This protocol therefore does not propose a generative device; it proposes a narrow, locked, validated device behavior governed by an on-premises repository-based LLM whose advisories are bounded, auditable, and held outside the robot vendor kinematic stack. The author's established record of LLM evaluation and trust over the period from August 2024 to June 2026 grounds this design, including a comparative code-generation evaluation of 16 proprietary versus open-source LLMs against the FDA Adverse Event Reporting System [24], a mobile pancreatic-cancer surgical-humanoid program with priority VVUQ [25], the 21 CFR part 312 Physical AI unification adaptation [6], and the verification-before-generation legislative framing [8].

Funding, co-investment, and the success likelihood of a definitive trial

A definitive, generalizable, and equitable Phase 2 answer is itself an outcome that can be made more or less likely by how the trial is resourced, and this protocol makes that mechanism explicit and governed. The Patient-Aligned Co-Investment Facility (PACIF) allows wealthier individuals connected to cancer patients to contribute ring-fenced philanthropic or private capital with the sole purpose of raising the trial's success likelihood. The defining safeguard is a capital firewall: contributors are structurally walled off from randomization, endpoint definition, outcome adjudication, statistical analysis, and publication, so that capital can never buy a favorable result. What capital can buy is limited to six operational success-likelihood levers, each of which acts on accrual, fidelity, retention, data quality, or oversight rather than on the answer. First, multi-site expansion to eight academic hepato-pancreato-biliary centers accelerates accrual and broadens generalizability. Second, Phase 0 compute funds ≥ 5000 simulations across ≥ 3 independent frameworks, lowering the sim-to-real gap (target < 1.5 mm and < 0.4 N) and raising the unified safety level to USL ≥ 8.0. Third, a robot fleet with redundancy increases throughput and uptime. Fourth, a Patient Access and Equity Fund covers travel, lodging, lost-wage, caregiver, and navigation costs, raising retention, which lowers dropout and thereby protects statistical power, while making enrollment more equitable and representative. Fifth, central infrastructure (BICR, central pathology, ctDNA, federated learning, and biostatistics) improves data quality and reduces bias. Sixth, independent oversight (a Data and Safety Monitoring Board, an independent safety monitor, a Physical AI Safety Review Committee, and an ethicist) protects safety and integrity. The facility is governed by the H.R. 9510 verification, validation, and uncertainty quantification (VVUQ) financial-data standard [8], disclosed under 21 CFR part 54 [3], and its results are deposited openly under CC BY 4.0. Table 3 maps each co-investment tranche to the operational lever it funds and to the firewall that keeps it away from the answer. In short, aligned capital raises power, fidelity, retention, equity, and generalizability without buying the answer.

Physical AI concerns addressed by this protocol

Nine recurring concerns are raised about Physical AI in patient care, and this protocol pairs each with a concrete, prespecified mitigation. The first eight concern the device and its governance; the ninth, new to this Phase 2 design, concerns financial influence and inequitable access and is answered by the capital firewall and the equity fund. Figure 4 maps the nine concerns to their mitigations, and Table 2 states each pairing in full.

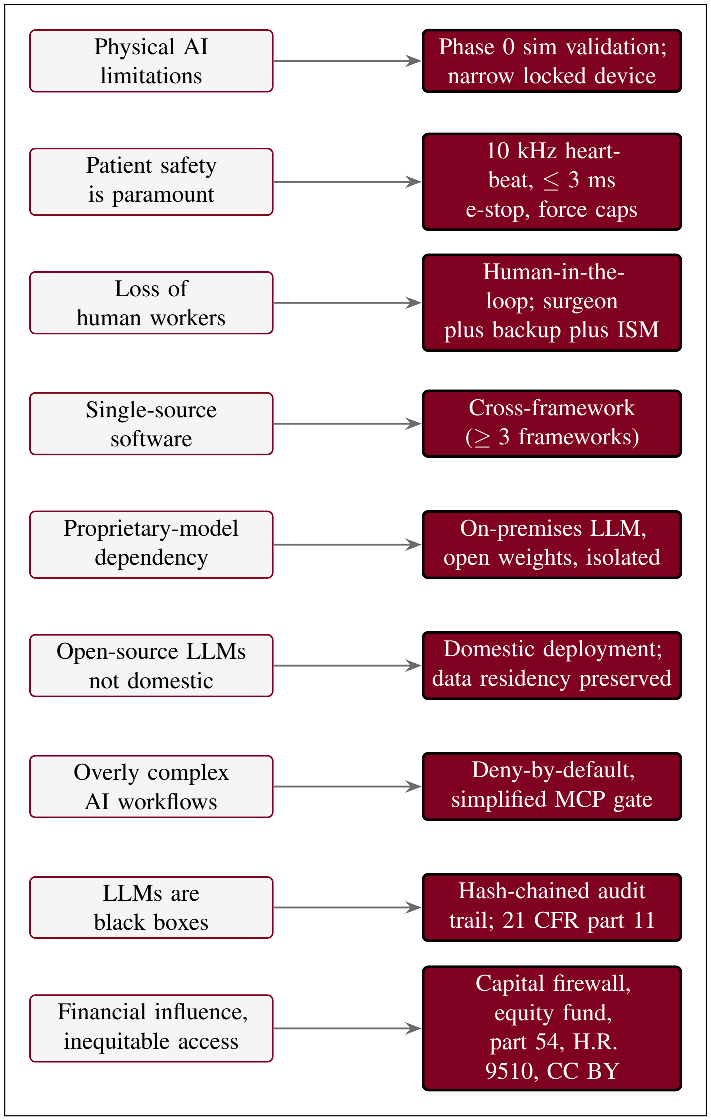

Figure 4. The mapping of all nine Physical AI concerns to protocol mitigations, the same nine pairs in Table 2. The original eight device concerns are paired with validated device behavior over open-endedness, layered hard safety limits, retained human roles, and multi-framework non-single-source validation; the ninth, financial influence and inequitable access, is answered by the capital firewall, the Patient Access and Equity Fund, 21 CFR part 54 disclosure, the H.R. 9510 VVUQ financial standard, and open CC BY deposition.

3.3 Risk/Benefit Assessment

3.3.1 Known Potential Risks

The known potential risks fall into three categories. Drug risks arise from daraxonrasib toxicity, including gastrointestinal effects (nausea, diarrhea, stomatitis), dermatologic and mucocutaneous events, hepatic enzyme elevations, and dose-limiting toxicities bounded at the established RP2D; perioperative exposure additionally raises a theoretical risk to wound and anastomotic healing if restart timing is sub-optimal. Device and robotic risks include device malfunction, an erroneous AI advisory, cyber-physical failure of the control or telemetry chain, and inadvertent vascular injury; these are the categories that motivate the significant-risk determination and the layered hard safety limits. Operative risks are those

Table 2: The nine Physical AI concerns and their protocol mitigations.

Concern	Protocol mitigation
Physical AI limitations	Phase 0 simulation validation over ≥ 5000 procedures and ≥ 3 independent frameworks; device behavior is narrow and locked, not generative, and operates only within validated envelopes.
Patient safety is paramount	Layered hard limits: a 10 kHz heartbeat broadcast bus, a ≤ 3 ms cross-arm emergency stop, ≤ 3 N per-arm and ≤ 18 N cumulative tip-force caps, and a five-vessel no-fly gate around the SMV and PV.
Loss of human workers	The human is retained in the loop: a credentialed surgeon, a backup operator, and an independent safety monitor (ISM) supervise every procedure, with the LLM acting as a bounded advisory oracle rather than a replacement.
Single-source software	Validation is cross-framework, with ≥ 3 independent frameworks required, so no single simulator or codebase is a single point of failure.
Proprietary-model dependency	The advisory uses an on-premises repository-based LLM with open weights, isolated from any single vendor stack, so the protocol is not captive to one proprietary model provider [24].
Non-domestic open-source LLMs	Deployment is domestic and on-premises, preserving data residency and avoiding reliance on non-domestic open-source model hosting; inference and records remain within the controlled environment.
Overly complex AI workflows	A deny-by-default posture with a deliberately simplified gate surface (a small set of Model Context Protocol servers) constrains the action space and reduces workflow complexity to an auditable minimum.
LLMs are black boxes	A hash-chained audit trail under 21 CFR part 11 records every advisory, command, and safety event, making each re-traceable to a deterministic seed and a specific commit, so the black box becomes traceable [6].
Financial influence and inequitable access	A capital firewall walls funders off from randomization, endpoints, adjudication, analysis, and publication; a Patient Access and Equity Fund supports equitable enrollment and retention; influence is disclosed under 21 CFR part 54, governed by the H.R. 9510 VVUQ financial standard, and all results are deposited openly under CC BY [8, 3].

Table 3: The patient-aligned co-investment tranches, the operational success-likelihood levers they fund, and the capital firewall.

Co-investment tranche	Operational lever funded	Effect on success likelihood; firewall note
Multi-site network (8 centers)	Eight academic HPB centers activated and supported	Faster accrual and broader generalizability of the PFS result; no influence on randomization or endpoints.
Phase 0 compute	$\geq$ 5000 simulations across $\geq$ 3 frameworks	Lower sim-to-real gap ($<$ 1.5 mm, $<$ 0.4 N) and USL $\geq$ 8.0, raising procedural fidelity; compute cannot alter the analysis plan.
Robot fleet plus redundancy	Device fleet, spares, and uptime engineering	Higher throughput and uptime, sustaining in-window accrual; no access to outcome adjudication.
Patient Access and Equity Fund	Travel, lodging, lost-wage, caregiver, and navigation support	Higher retention that lowers dropout and protects statistical power, with more equitable and representative enrollment; funders see no patient-level outcomes.
Central review and biomarker core	BICR, central pathology, ctDNA, and federated learning	Improved data quality and reduced bias; the core is independent of contributors and of the analysis team.
Independent oversight	DSMB, ISM, Physical AI Safety Review Committee, ethicist	Stronger safety and integrity monitoring; oversight bodies are insulated from funders.

Capital firewall: contributors to any tranche cannot influence randomization, endpoint definition, outcome adjudication, statistical analysis, or publication; capital buys only operational success-likelihood, never a favorable result. Influence is disclosed under 21 CFR part 54 and governed by the H.R. 9510 VVUQ financial-data standard [8, 3].

intrinsic to the Whipple procedure: postoperative pancreatic fistula (graded by ISGPS [11]), hemorrhage, delayed gastric emptying, intra-abdominal infection and sepsis, and the need for conversion to open surgery. Because the design is randomized, patients in the control arm (Arm B) also carry the operative and modified-FOLFIRINOX toxicity risks intrinsic to the standard of care [15]. The contemporary robotic baseline quantifies the residual magnitude of these operative risks even in expert hands [10].

3.3.2 Known Potential Benefits

The known potential benefits are directed at the specific failure modes above. A precise, in-window R0 resection is the principal oncologic benefit, addressing the resection-window collapse and the margin-positivity that drive shorter survival. Layered hard safety limits with a full hash-chained audit trail provide a benefit not available in manual surgery: every machine action is bounded and re-traceable, supporting both intraoperative safety and post hoc verification. Optimized drug-restart timing, advised at T+7, T+14, or T+21 days and keyed to fistula status and drug trough, aims to restore systemic therapy at the earliest safe point and thereby suppress micrometastatic outgrowth. Beyond the individual procedure, the Patient Access and Equity Fund confers a direct participant benefit, lowering the travel, lodging, lost-wage, and caregiver burdens that drive attrition and inequitable enrollment, which improves both retention and the representativeness of the evidence. For a low-survival population with few alternatives, even incremental gains in margin status, complication avoidance, therapy continuity, and equitable access are clinically meaningful.

3.3.3 Assessment of Potential Risks and Benefits

Figure 5 frames the benefit-risk weighing for this population, now as a randomized comparison with an internal control. Against a baseline of under-13-percent five-year survival and few alternatives, the novel risks (device malfunction, advisory error, cyber-physical failure) are weighed against the benefits (precise in-window R0 resection, layered hard safety limits with a full audit trail, optimized drug-restart timing, and the access-fund gains in retention and equity). For this low-survival, limited-alternative population, the benefit-risk balance is favorable: the potential benefits are likely to outweigh the risks, and the assessment is consistent with the expanded-access ethic that permits investigational treatment for serious or immediately life-threatening conditions when the potential benefit justifies the potential risk (21 CFR §312.84).

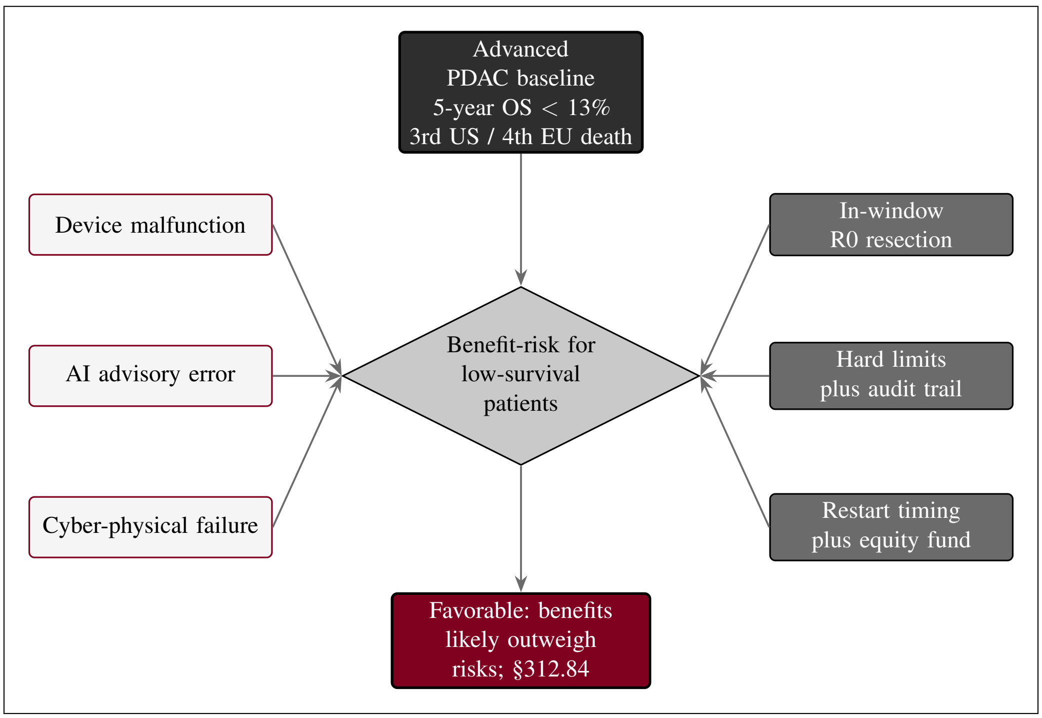

Figure 5. Risk-benefit framework for the randomized Phase 2 comparison in advanced PDAC. Against a baseline of under-13-percent five-year survival and few alternatives, the novel risks (device malfunction, advisory error, cyber-physical failure) are weighed against the benefits (precise in-window R0 resection, layered hard safety limits with a full audit trail, optimized drug timing, and the access-fund gains in retention and equity). For this low-survival, limited-alternative population the benefit-risk balance is favorable.

Finally, Figure 6 makes the co-investment logic explicit and shows where the firewall sits. Aligned capital and the PACIF supply only the operational levers; the capital firewall blocks every path from a contributor to the endpoints, their adjudication, or the analysis, so the levers raise the success likelihood of a definitive answer while the integrity of that answer remains structurally protected.

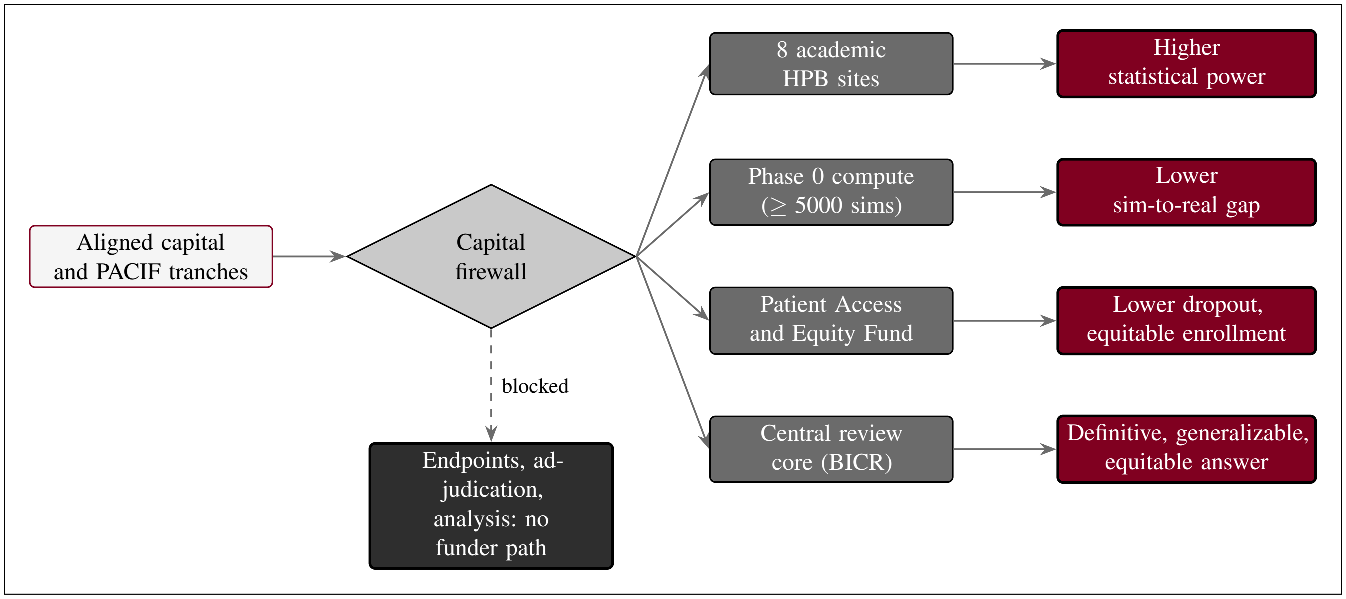

Figure 6. The co-investment-to-success-likelihood pathway. Aligned capital and the PACIF flow only through the capital firewall into operational levers (eight sites, Phase 0 compute, the Patient Access and Equity Fund, and the central review core), which raise statistical power, lower the sim-to-real gap, lower dropout, broaden equitable enrollment, and so raise the probability of a definitive, generalizable, equitable answer. The dashed blocked edge shows that the firewall admits no path from funders to the endpoints, their adjudication, or the analysis: capital raises operational success likelihood while the integrity of the result is structurally protected [8, 3].

4 Objectives and Endpoints

This study pairs each objective with a prospectively defined, measurable endpoint and a scientific justification, presented in the standard three-column form. Unlike the single-arm first-in-human predicate [1], this protocol is a randomized controlled efficacy study that directly compares Arm A (perioperative daraxonrasib (RMC-6236) at the recommended Phase 2 dose (RP2D) plus on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple), PancreSpeed II) against Arm B (modified FOLFIRINOX [15] plus institutional-standard high-volume pancreaticoduodenectomy). The objective set is therefore organized around a single confirmatory efficacy question, a fixed-sequence hierarchy of key secondary questions that share the study-wide error budget, a set of estimation-only secondary measures, and a tier of exploratory measures. Table 4 is the governing objectives-endpoints table; it drives the analysis plan in §9 and the safety rules in §6 and §7.

Abbreviations in Trial Protocol: BICR, blinded independent central review; ctDNA, circulating tumor DNA; EBL, estimated blood loss; HR, hazard ratio; ISGPS, International Study Group on Pancreatic Surgery; LLM, large language model; LOS, length of stay; MPR, major pathologic response; OS, overall survival; PDAC, pancreatic ductal adenocarcinoma; PFS, progression-free survival; QoL, quality of life; RECIST, Response Evaluation Criteria in Solid Tumors; RP2D, recommended Phase 2 dose. The internal control is Arm B; the 2025 Dutch nationwide cohort of 1000 robotic pancreaticoduodenectomies [10] is retained only as an external descriptive benchmark.

Figure 7 renders the objective-to-endpoint hierarchy that organizes Table 4, showing how the single primary efficacy comparison of PFS between Arm A and Arm B sits above the fixed-sequence key secondary family, which in turn sits above the estimation-only secondary measures and the exploratory measures.

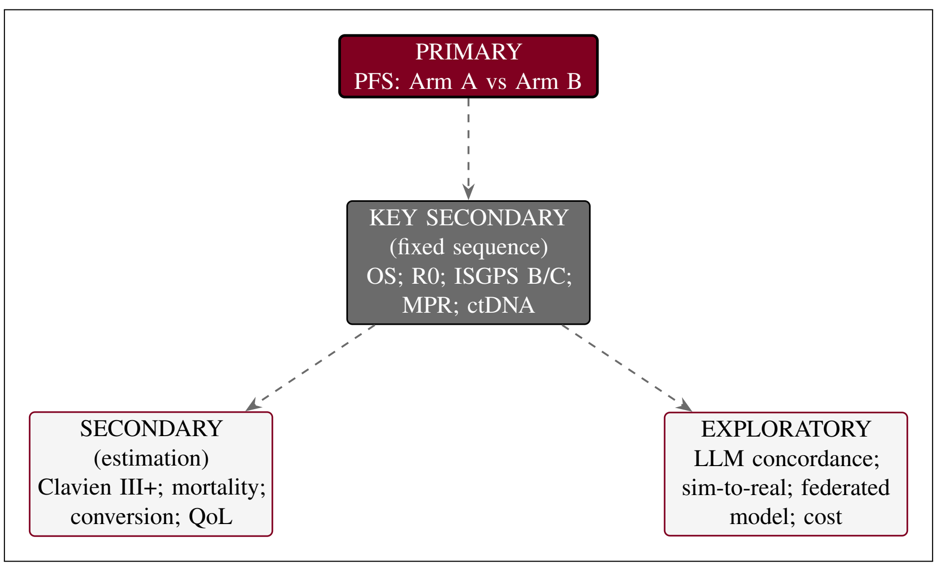

Figure 7. Objective-to-endpoint hierarchy for the randomized comparison. The primary efficacy endpoint (progression-free survival, Arm A versus Arm B) sits above the fixed-sequence key secondary family (overall survival, R0 resection rate, ISGPS grade B/C fistula, major pathologic response, and week-12 KRAS ctDNA clearance), which sits above the estimation-only secondary measures (Clavien-Dindo grade III+, 90-day mortality, conversion rate, and quality of life) and the exploratory measures (LLM advisory concordance, the sim-to-real gap, federated model performance, and cost-effectiveness). Dashed edges separate the tiers.

Table 4: Objectives, endpoints, and justifications. The primary objective is confirmatory; the key secondary objectives are tested in a fixed-sequence hierarchy that controls the family-wise error rate; secondary objectives are estimation only; exploratory objectives are hypothesis-generating.

Objective	Endpoint	Justification
Primary		
Compare progression-free survival (PFS) between Arm A and Arm B.	PFS, defined as time from randomization to RECIST 1.1 progression or death from any cause, assessed by the investigator with blinded independent central review (BICR) as a sensitivity analysis.	PFS is the standard randomized Phase 2 efficacy anchor in resectable and borderline-resectable PDAC; a hazard ratio favoring Arm A supports progression to a Phase 3 confirmatory trial.
Key Secondary (hierarchical, fixed sequence)		
1. Compare overall survival (OS) between arms.	OS, time from randomization to death from any cause.	OS is the definitive measure of clinical benefit and the first gatekeeper after PFS in the testing hierarchy.
2. Compare the R0 resection rate between arms.	Proportion with a microscopically margin-negative (R0) resection on central masked pathology review.	R0 status is the strongest modifiable surgical determinant of survival in resectable PDAC.
3. Compare clinically relevant postoperative pancreatic fistula between arms.	ISGPS grade B or C postoperative pancreatic fistula rate [11].	Grade B/C fistula is the leading driver of post-Whipple morbidity and a key safety-quality discriminator between the two operations.
4. Compare major pathologic response between arms.	Major pathologic response (MPR) rate on central masked pathology review.	MPR is a validated surrogate of neoadjuvant treatment effect that links the systemic regimen to the resection specimen.
5. Compare molecular response between arms.	KRAS circulating tumor DNA (ctDNA) clearance at week 12 [12].	Week-12 ctDNA clearance is an early, minimally invasive molecular readout of disease control that complements the imaging-based primary endpoint.
Secondary (estimation)		
Estimate major postoperative complications by arm.	Clavien-Dindo grade III or higher complication rate through 90 days; external comparator 41.3% [10].	Clavien-Dindo grade III+ is the validated severity threshold for complications requiring procedural, endoscopic, or intensive-care management.
Estimate early postoperative mortality by arm.	90-day all-cause mortality.	Ninety-day mortality is the standard short-term survival safety measure for major hepatopancreatobiliary surgery.
Estimate the conversion pathway by arm.	Unplanned conversion rate from the robotic system to an open or conventional approach; external comparator 10.1% [10].	Conversion rate contextualizes device performance against established robotic practice.
Estimate operative burden and recovery by arm.	Operative time, estimated blood loss (EBL), and length of stay (LOS).	These operative and recovery measures characterize the procedural footprint of each operation.
Estimate quality of life by arm.	EORTC QLQ-C30 and PAN26 scores over follow-up [16]; 12- and 18-month PFS rates and 24-month OS rate.	Patient-reported quality of life and landmark survival rates frame the patient-centered value of each regimen.
Exploratory		
Characterize the LLM advisory layer.	Concordance between the LLM advisory output and the independent hard-coded safety gate, as a proportion of advisories.	Advisory-to-gate concordance characterizes the second-opinion oracle and informs autonomy-graduation decisions in later phases.
Characterize simulation fidelity	Sim-to-real gap in trajectory (mm) and tip	These measures validate the upgraded

4.1 Primary Objective and Endpoint

The single primary objective is to compare progression-free survival between Arm A and Arm B. The primary endpoint is PFS, defined as the time from randomization to the first documented RECIST 1.1 disease progression or death from any cause, whichever occurs first. The primary analysis is investigator-assessed, with BICR applied as a prespecified sensitivity analysis to guard against evaluation bias in an open-label design. PFS is the established randomized Phase 2 efficacy anchor in resectable and borderline-resectable PDAC: it is measurable within the trial horizon, it integrates the systemic and surgical effects of each arm, and a hazard ratio favoring Arm A provides the confirmatory signal needed to justify a Phase 3 trial. The study is sized for approximately 140 PFS events to provide 85% power at a two-sided significance level of 0.05 to detect a hazard ratio of 0.60 (median PFS 14.0 months in Arm B versus 23.3 months in Arm A), with a single interim analysis for efficacy and futility at approximately 60% of events governed by an O'Brien-Fleming spending function [13, 26] (§9).

4.2 Key Secondary Objectives and Endpoints

The key secondary objectives are tested in a single fixed-sequence (hierarchical) gatekeeping procedure that controls the family-wise type I error rate at a two-sided 0.05 level. Each hypothesis is tested at the full 0.05 level only if the primary PFS comparison and every preceding key secondary hypothesis have reached statistical significance; the first non-significant result closes the sequence, and all subsequent comparisons are reported as descriptive only. The fixed order is: (1) overall survival; (2) R0 resection rate; (3) ISGPS grade B/C postoperative pancreatic fistula rate [11]; (4) major pathologic response rate; and (5) week-12 KRAS ctDNA clearance [12]. Overall survival is placed first as the definitive measure of clinical benefit. R0 resection rate and MPR are assessed on central masked pathology review, and the ISGPS fistula grade is adjudicated centrally, so that each surgical-quality comparison is protected from unmasking in the open-label setting. Week-12 ctDNA clearance provides an early molecular corroboration of the imaging-based primary endpoint. This single ordered sequence keeps the confirmatory inference for the entire key secondary family within one shared error budget.

4.3 Secondary Objectives and Endpoints

The secondary objectives are estimation only; each is summarized with a point estimate and a two-sided 95% confidence interval (for between-arm contrasts, a difference or hazard ratio with its interval), and none is part of the confirmatory hierarchy. They comprise the **Clavien-Dindo grade III or higher complication rate** through 90 days, for which the 2025 Dutch nationwide robotic cohort reported 41.3% [10]; **90-day all-cause mortality**; the **unplanned conversion rate** from the robotic system to an open or conventional approach, with a Dutch external comparator of 10.1% [10]; **operative time, estimated blood loss, and length of stay**; and **quality of life** measured by the EORTC QLQ-C30 and PAN26 instruments [16], together with the 12- and 18-month PFS rates and the 24-month OS rate. Because these measures sit outside the testing hierarchy, the Dutch values are interpreted as external descriptive benchmarks and the between-arm contrasts are interpreted as estimates rather than as confirmatory findings; the internal Arm B comparator remains the primary reference frame.

4.4 Exploratory Objectives and Endpoints

The exploratory objectives generate hypotheses for later-phase study and are explicitly non-confirmatory; no exploratory analysis is used to make a go or no-go decision on the intervention, and all are reported with that caveat. **LLM advisory-to-gate concordance** is the proportion of LLM advisory outputs that agree with the independent hard-coded safety gate, quantifying the reliability of the second-opinion oracle and informing autonomy-graduation decisions reserved for later phases

(§4). The **sim-to-real gap** is the difference between the Phase 0 simulation prediction and the intra-operative record, reported as the trajectory gap in millimeters and the tip-force gap in newtons against the upgraded Phase 0 thresholds, and is reported alongside **federated-learning model performance** and **ctDNA dynamics** pooled across the eight sites to characterize the multicenter fleet. The **health-economic and cost-effectiveness** outcomes are linked to the capital-firewalled co-investment design and inform the equitable-access rationale. Because these measures are descriptive, they are summarized with point estimates and dispersion only, without confirmatory significance testing.

5 Study Design

5.1 Overall Design

This is a prospective, multicenter, randomized, parallel-group, controlled, open-label study with blinded independent central review (BICR), conducted across eight academic high-volume hepatopancreatobiliary (HPB) centers. A total of $n = 220$ participants are randomized 1:1 (110 per arm). Arm A receives perioperative daraxonrasib (RMC-6236) at the recommended Phase 2 dose (RP2D) of 300 mg orally once daily plus the on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) system (PancreSpeed II). Arm B receives modified FOLFIRINOX [15] plus institutional-standard high-volume pancreaticoduodenectomy that is not LLM-directed. Allocation uses central permuted-block randomization 1:1, stratified by resectability (resectable versus borderline-resectable), KRAS allele (G12D versus other G12), neoadjuvant therapy (yes versus no), and site. The study is conducted under a combined IND/IDE submission with a single integrated protocol, a single informed consent carrying the Physical AI opt-out, and a single multicenter safety governance structure.

The device operates at Stage 2 supervised semiautonomy as a Class II collaborative system under continuous human oversight with an immediate manual override; full autonomy is prohibited under 21 CFR §312.21(e), and any expanded supervised-autonomous subtask requires independent safety review and a Unified Safety Level (USL) of at least 8.0. A mandatory upgraded Phase 0 simulation-validation gate must be passed at each site before any patient-facing robotic activity (§4.3). The eight-site, powered, equitable design is enabled by the Patient Access and Equity Fund (the Patient-Aligned Co-Investment Facility, PACIF) held behind a capital firewall, so that funders cannot touch randomization, endpoint definition, outcome adjudication, or analysis [8, 3]. Figure 8 presents the randomization and multicenter design.

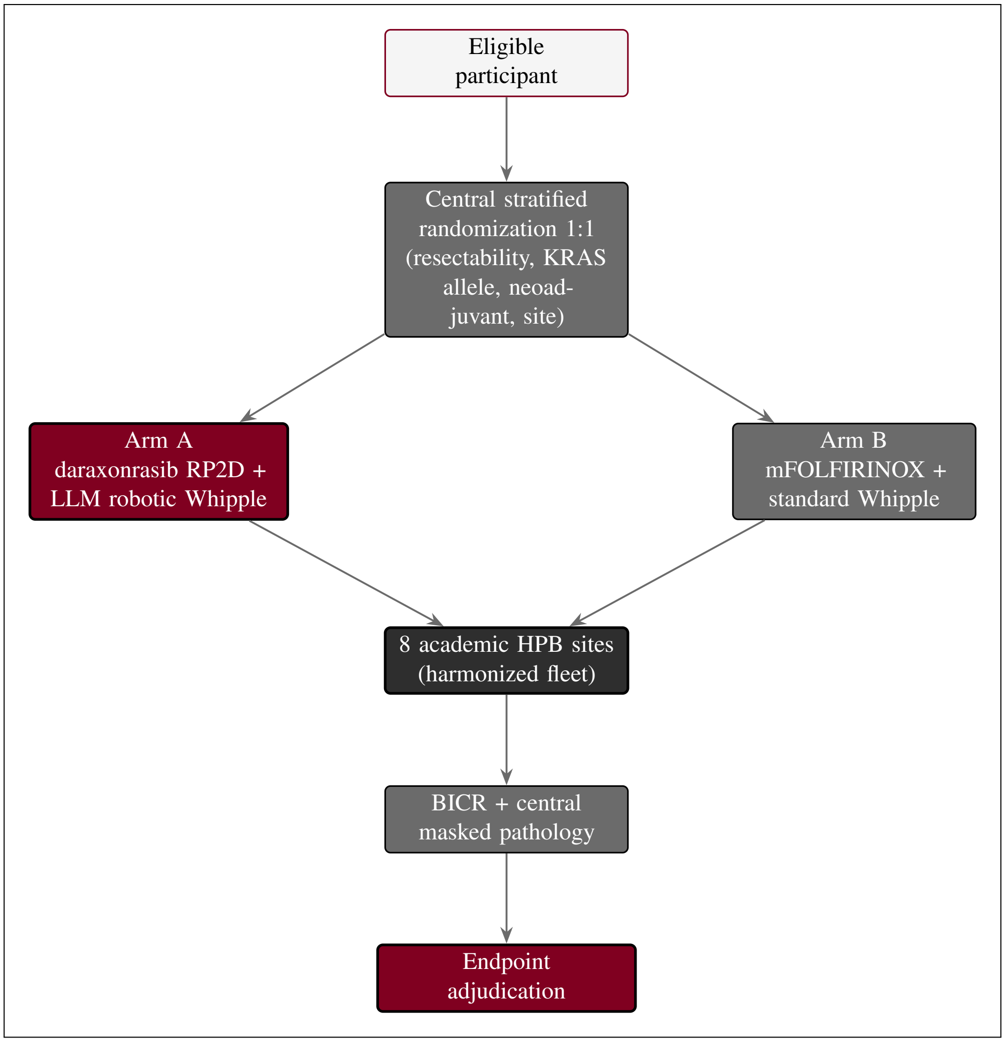

Figure 8. Randomization and multicenter design. Eligible participants undergo central permuted-block randomization 1:1, stratified by resectability, KRAS allele, neoadjuvant therapy, and site, to Arm A (daraxonrasib at the RP2D plus the LLM-directed robotic Whipple) or Arm B (modified FOLFIRINOX plus standard high-volume Whipple). Both arms are delivered across the eight academic HPB sites operating as a harmonized fleet, and all outcomes flow through blinded independent central review and central masked pathology into blinded endpoint adjudication.

5.2 Scientific Rationale for Study Design

A randomized, controlled, parallel-group design is now the scientifically and ethically appropriate next step, inverting the logic of the single-arm predicate. The first-in-human study was single-arm precisely because the early safety profile, the RP2D, and the technical feasibility of the combined drug-device intervention were unknown, and randomizing a vulnerable population before that evidence existed could not be justified. The Phase 1 predicate [1] has now established all three, so genuine clinical equipoise exists between the investigational regimen and institutional-standard care, and a randomized controlled comparison is the appropriate and ethical way to test comparative efficacy. The trial follows the CONSORT framework for reporting a parallel-group randomized design [27].

Open-label conduct is unavoidable because a surgical-system intervention cannot be masked from the operating team or the participant. Bias control is therefore concentrated downstream of the intervention: the primary PFS endpoint is supported by BICR of imaging, the R0 and MPR endpoints are read on central masked pathology, and all confirmatory endpoints pass through blinded endpoint adjudication, so that the open-label delivery does not propagate into outcome ascertainment. The internal control, Arm B, replaces the external Dutch benchmark used in the predicate as the primary comparator [10]; the Dutch cohort is retained only as an external descriptive reference. The study population remains vulnerable, with a total five-year survival below 13% [9], so the Patient Access and Equity Fund and per-cohort DSMB oversight are maintained to protect participants throughout the randomized comparison.

5.3 Justification for Dose and Device Readiness

Daraxonrasib at the established RP2D. Daraxonrasib is administered at the RP2D of 300 mg orally once daily established in the Phase 1 predicate and anchored to the RASolute 302 pan-KRAS program [1, 14]. Because the RP2D is already established, there is no 3+3 dose escalation in this Phase 2 study; every participant in Arm A receives the same fixed dose, and the trial tests comparative efficacy rather than dose-finding.

Device "dose" analogue and the upgraded Phase 0 gate. The device has no pharmacologic dose; its dose analogue is the readiness evidence required before any patient-facing activity, escalated only by passing the mandatory upgraded Phase 0 simulation-validation gate. Relative to the predicate, the gate is upgraded to require a USL rating of at least 8.0, at least 5000 simulated procedures reproduced across at least three independent simulation frameworks, and a tightened sim-to-real fidelity of a trajectory gap below 1.5 mm and a tip-force gap below 0.4 N, together with per-site qualification and multicenter fleet harmonization so that every device instance behaves identically across the eight centers. These preconditions operate with the intra-operative hard caps (per-arm tip-force caps of ≤ 3 N, a cumulative cross-arm cap of ≤ 18 N, a cross-arm emergency-stop budget of ≤ 3 ms, and a five-vessel no-fly gate around the major peripancreatic vessels) [4, 6]. The expanded Phase 0 compute required to meet these thresholds is funded by the co-investment facility behind the capital firewall, which purchases operational capacity only and never scientific outcomes. Figure 9 presents the staged-autonomy model for Phase 2.

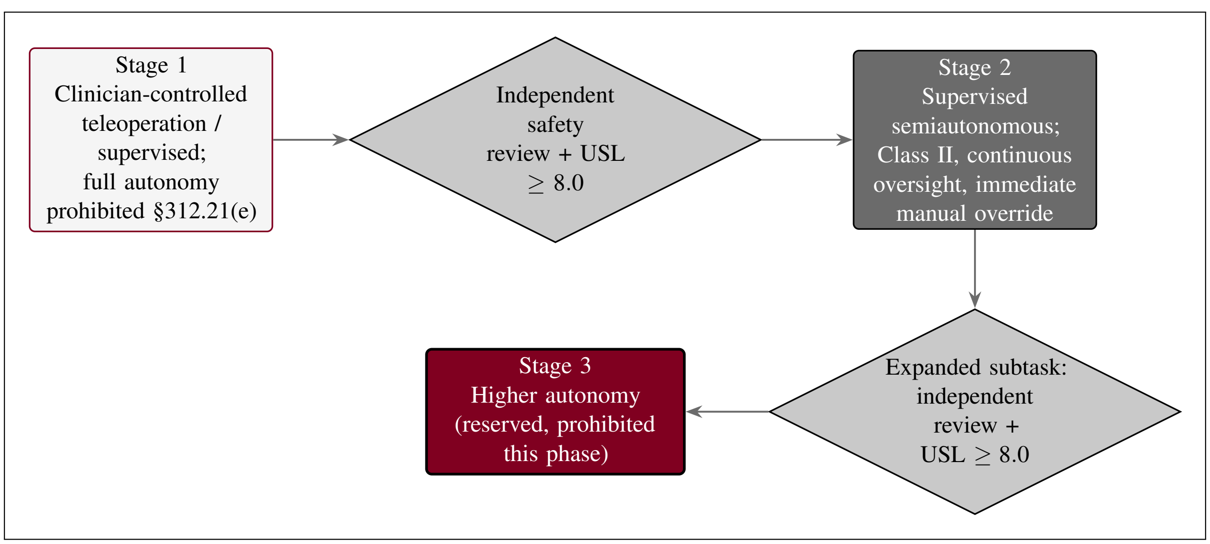

Figure 9. Phase-graduated staged-autonomy model for Phase 2. The operative system runs at Stage 2 supervised semiautonomy (Class II, continuous oversight with immediate manual override). Any expanded supervised-autonomous subtask requires an independent safety review and a Unified Safety Level of at least 8.0; full autonomy (Stage 3) remains reserved and prohibited at this phase under 21 CFR §312.21(e).

5.4 End of Study Definition

An individual participant is considered to have completed the study at the last visit specified in the Schedule of Activities (§1.3), which is the 24-month overall-survival assessment, or at the participant's death, documented disease progression precluding further follow-up, or withdrawal of consent, whichever occurs first. The end of the study as a whole is defined as the date of the last scheduled PFS or OS assessment for the last enrolled participant together with the primary-analysis database lock, which occurs after approximately 140 PFS events have accrued. After that date, no further study-specific procedures are performed. Participants who discontinue the investigational intervention for a protocol-defined reason (§7) remain in long-term follow-up for survival to the 24-month endpoint unless they withdraw consent for follow-up.

6 Study Population

The study population comprises adults with histologically confirmed, resectable or borderline-resectable, KRAS G12-mutated pancreatic ductal adenocarcinoma (PDAC) who are eligible for daraxonrasib under broad pan-KRAS criteria and whose anatomy is compatible with the on-premises eight-arm robotic system for those who may receive the experimental arm. This is a multicenter, randomized, controlled, open-label study conducted across eight academic high-volume hepatopancreatobiliary (HPB) centers. Target enrollment is approximately 245 participants screened to yield 220 randomized 1:1 (110 per arm), where the modest screen-to-randomization margin allows for up to 10 percent non-evaluability while preserving the planned statistical power (§9). Participants are allocated to Arm A (experimental: perioperative daraxonrasib at the recommended Phase 2 dose plus the LLM-directed robotic Whipple) or Arm B (active control: modified FOLFIRINOX [15] plus institutional-standard high-volume pancreaticoduodenectomy). Figure 10 presents the randomized two-arm CONSORT participant flow from screening through allocation to the analysis populations.

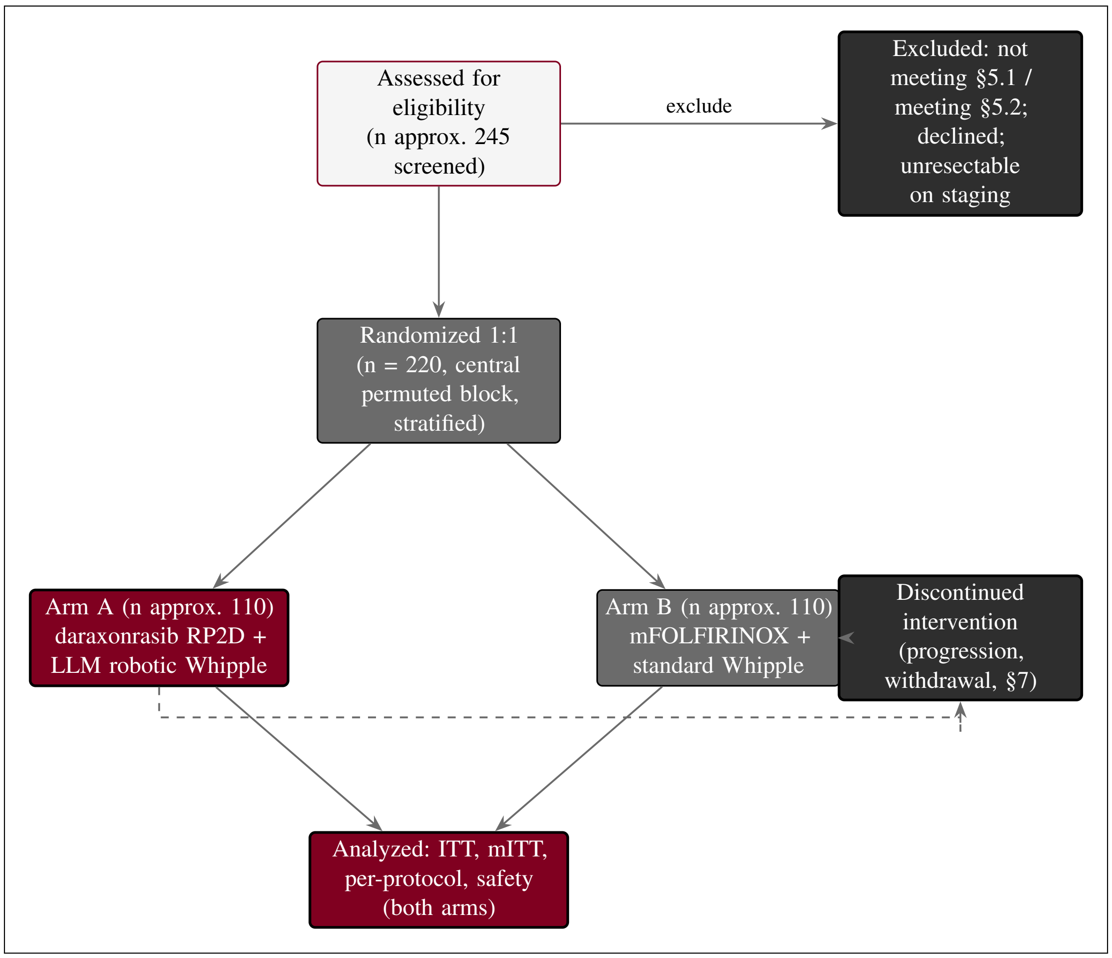

Figure 10. Randomized two-arm CONSORT participant flow. Approximately 245 individuals are assessed for eligibility; after exclusions, 220 are randomized 1:1 by central permuted-block randomization stratified by resectability, KRAS allele, neoadjuvant therapy, and site, to Arm A (daraxonrasib at the RP2D plus the LLM-directed robotic Whipple, n approximately 110) or Arm B (modified FOLFIRINOX plus standard high-volume Whipple, n approximately 110). Both arms feed the intention-to-treat, modified intention-to-treat, per-protocol, and safety analysis populations, with a protocol-defined discontinuation branch.

6.1 Inclusion Criteria

To be eligible to participate in this study, an individual must meet all of the following criteria.

1. Signed and dated informed consent obtained before any study-specific procedure, including the participant's explicit election or declination of the Physical AI opt-out, which governs use of the LLM-directed advisory layer for those allocated to the experimental arm (§8).
2. Age $\geq$ 18 years at the time of consent.
3. Histologically or cytologically confirmed PDAC that is resectable or borderline-resectable on multidisciplinary radiologic review.
4. A documented KRAS G12 mutation (for example G12D, G12V, or G12R) on a validated tumor genotyping assay.
5. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1.
6. Adequate organ and marrow function on screening laboratories, including hematologic, hepatic, and renal indices sufficient to undergo major pancreaticobiliary surgery and to receive either study regimen.
7. A measured baseline CA 19-9 value (no threshold imposed; value recorded for serial assessment).
8. Eligible for daraxonrasib under the broad pan-KRAS criteria of RASolute 302 [14].
9. Tumor and vascular anatomy compatible with the eight-arm robotic system on the preoperative imaging review, including port placement and working volume adequate for operative tasks, for participants who may receive Arm A.

The synthetic reference exemplar PAT-PDAC-0002 illustrates a representative eligible participant: a 66-year-old with a 3.2 cm head-of-pancreas tumor abutting the superior mesenteric vein, KRAS G12D, microsatellite stable, CA 19-9 elevated, ECOG performance status 1, after a completed neoadjuvant modified FOLFIRINOX course [15]. This exemplar is synthetic; it is used for design illustration and does not represent an enrolled individual.

6.2 Exclusion Criteria

An individual who meets any one of the following criteria is excluded from participation.

1. Distant metastatic disease on staging.
2. Vascular involvement that precludes a margin-negative (R0) resection.
3. Prior pancreatic resection or prior surgery that precludes the planned reconstruction.
4. Uncontrolled intercurrent illness or comorbidity (for example uncontrolled cardiac, pulmonary, hepatic, renal, or active infectious disease) that would make major surgery or either study regimen unsafe.
5. Pregnancy or lactation, or unwillingness to use adequate contraception during the study if applicable.
6. Known contraindication or hypersensitivity to daraxonrasib or to modified FOLFIRINOX or any of their required components, or a concomitant medication interaction that cannot be safely managed.
7. Inability to provide informed consent or to comply with the Schedule of Activities.

Participant selection is equitable. Eligibility is determined solely by the scientific and safety criteria above, and no eligible individual is excluded on the basis of sex, gender, race, ethnicity, socioeconomic status, or insurance status. Limited English proficiency is not an exclusion criterion; qualified interpreters and translated consent materials are provided so that individuals with limited English proficiency can be fully informed and can participate on the same basis as English-speaking individuals. The Patient Access and Equity Fund (the Patient-Aligned Co-Investment Facility, PACIF) removes travel, lodging, lost-wage, caregiver, and navigation barriers behind a capital firewall, so that representative enrollment across the eight centers is not limited by a participant's financial circumstances and so that funders cannot influence randomization or outcomes [8, 3].

6.3 Lifestyle Considerations

Perioperative lifestyle restrictions apply and are reviewed at consent and at the baseline visit. Participants follow a standardized perioperative nutrition plan, including the institutional preoperative fasting protocol and a staged postoperative diet advancement keyed to recovery of gastrointestinal function and, in Arm A, to fistula status. Alcohol is to be avoided during the perioperative period and during any interval of active daraxonrasib dosing, both to limit hepatotoxic burden and to avoid confounding the safety assessment. Tobacco cessation is strongly advised and is supported with referral, given its effect on wound healing and pulmonary risk. No restriction on routine physical activity is imposed beyond standard postoperative surgical guidance. There are no study-specific dietary caffeine or grapefruit restrictions beyond those required by the daraxonrasib labeling and the concomitant-medication rules in §6.

6.4 Screen Failures

A screen failure is an individual who consents to participate but is not subsequently enrolled and randomized because one or more eligibility criteria are not met. For each screen failure, a minimal CONSORT screen-failure data set is recorded: demography, the specific criterion or criteria that were not met (the screen-failure detail), the eligibility determination, and any serious adverse event that occurred between consent and the determination. No other study procedures are performed after a screen-failure determination. An individual who fails screening may be rescreened if the disqualifying condition was transient and is expected to resolve (for example a correctable laboratory abnormality); a rescreened individual retains the same participant number, and the rescreening is documented so that the screening-to-randomization accounting in Figure 10 remains complete.

6.5 Strategies for Recruitment and Retention

Recruitment is conducted across eight academic high-volume HPB centers operating as a coordinated network. Eligible individuals are identified at each site's tumor-board review and at surgical and medical-oncology clinics, and the study is described to them by a member of the research team who is not the treating surgeon, to reduce the potential for undue influence. The planned accrual rate is approximately 10 to 14 participants per month network-wide over an enrollment period of approximately 30 months, sufficient to randomize 220 participants from approximately 245 screened.

Retention through the 24-month overall-survival window is supported by structured follow-up with appointment reminders, flexible scheduling, and coordination with referring providers for participants who travel. Retention is further protected by the Patient Access and Equity Fund, which provides travel, lodging, lost-wage, caregiver support, and navigation behind a capital firewall: by removing the financial and logistical barriers that drive differential dropout, the fund lowers attrition and thereby protects the statistical power of the comparison and supports equitable, representative enrollment, while the firewall ensures that funders cannot influence randomization, endpoint definition, outcome adjudication, or analysis [8, 3]. Because the population is vulnerable, additional safeguards consistent with Office for Human Research Protections (OHRP) guidance are applied: enrollment uses an independent consent monitor when indicated, the consent process explicitly addresses therapeutic misconception about the LLM-directed robotic system and the voluntariness of the Physical AI opt-out, and participants are reminded at each contact that they may withdraw at any time without affecting their clinical care. Compensation is limited to reimbursement of documented study-related travel, lodging, lost-wage, caregiver, and related expenses provided through the fund; no payment is offered for undergoing the surgical procedure or the investigational drug, so that compensation does not function as an undue inducement.

7 Study Intervention

This study compares two randomized arms under a single integrated protocol. Arm A, the experimental arm, combines an investigational drug, perioperative daraxonrasib (RMC-6236) administered at the recommended Phase 2 dose (RP2D), with an investigational device, the on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) system (PancreSpeed II). Arm B, the active control, is modified FOLFIRINOX [15] plus institutional-standard high-volume pancreaticoduodenectomy that is not LLM-directed. This section describes both arms, their administration, their preparation and accountability, the measures taken to minimize bias, the compliance machinery for each, and the permitted and prohibited concomitant therapy. In Arm A the drug and device are coupled at one decision point only: the perioperative pause-and-restart advisory for the drug is informed by the operative and pathology results of the device procedure (§6.1.2).

7.1 Study Intervention Administration

7.1.1 Study Intervention Description

Experimental arm drug: daraxonrasib (RMC-6236). Daraxonrasib is an orally bioavailable, RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor that binds the active, guanosine triphosphate bound state of mutant RAS in complex with cyclophilin A, suppressing downstream effector signaling across the common KRAS oncoprotein variants found in PDAC. The agent received FDA Breakthrough Therapy designation in June 2025 for previously treated metastatic PDAC [17], and its dose is anchored to the pan-KRAS RASolute 302 program, in which 300 mg once daily is the established RP2D [14, 1]. In Arm A daraxonrasib is administered orally, once daily, in a perioperative schedule: a defined pre-operative course, a protocol-mandated pause spanning the operative window, and a postoperative restart whose timing is governed by the advisory described in §6.1.2. The drug is investigational in this perioperative, curative-intent surgical context and for the drug-device use here.

Control arm: modified FOLFIRINOX plus standard pancreaticoduodenectomy. Arm B delivers institutional-standard care. The systemic component is modified FOLFIRINOX, the multi-agent regimen established in the resectable and adjuvant PDAC setting [15], administered on its standard schedule with the institution's supportive-care practice. The surgical component is a high-volume pancreaticoduodenectomy performed by the institutional standard, robotic or open, that is not LLM-directed and does not use the eight-arm investigational platform. Arm B thus represents the contemporary standard of care against which the experimental drug-device regimen is compared.

Experimental device: on-premises LLM-directed eight-arm Whipple (PancreSpeed II). The investigational device is an eight-arm parallel coelomic surgical platform in which an on-premises repository-pinned LLM acts as a second-opinion advisory oracle held strictly outside the robot vendor kinematic stack, never as an autonomous controller. The platform carries 56 mechanical degrees of freedom (8×7 per-arm degrees of freedom) and a 640-channel mixed-rate sensor stack (80 channels per arm). Force channels sample at 100 kHz; all remaining channels sample at the 10 kHz command rate. Positioning accuracy is 0.05 mm root mean square at a peak tip velocity of 1,200 mm/s. The hard cyber-physical safety envelope comprises a per-arm tip-force cap of ≤ 3 N, a cumulative cross-arm tip-force cap of ≤ 18 N enforced at the heartbeat boundary, a 10 kHz heartbeat coordination bus broadcasting a 64-byte frame on a $100\,\mu$s per-arm watchdog deadline, an emergency arm park within $50\,\mu$s of a missed or out-of-parameter frame, and a cross-arm emergency stop (E-stop) that halts the full platform in ≤ 3 ms; a software-independent hardware E-stop backs the software chain and acts within ≤ 500 ms. A five-vessel no-fly gate enforces standoff distances around the major peripancreatic vessels (1.0 mm for the superior mesenteric vein and portal vein, 1.5 mm for the hepatic artery, celiac axis, and superior mesenteric artery, a 3.0 mm soft margin, and a 5.0 mm hard-stop margin). The platform is multicenter fleet-harmonized so that every instance behaves identically across the eight centers. In this Phase 2 protocol the device operates at Stage 2 supervised semiautonomy under continuous human oversight with immediate manual override; full autonomy is prohibited consistent with 21 CFR §312.21(e). The advisory control loop and platform architecture are shown in Figure 11 and Figure 12.

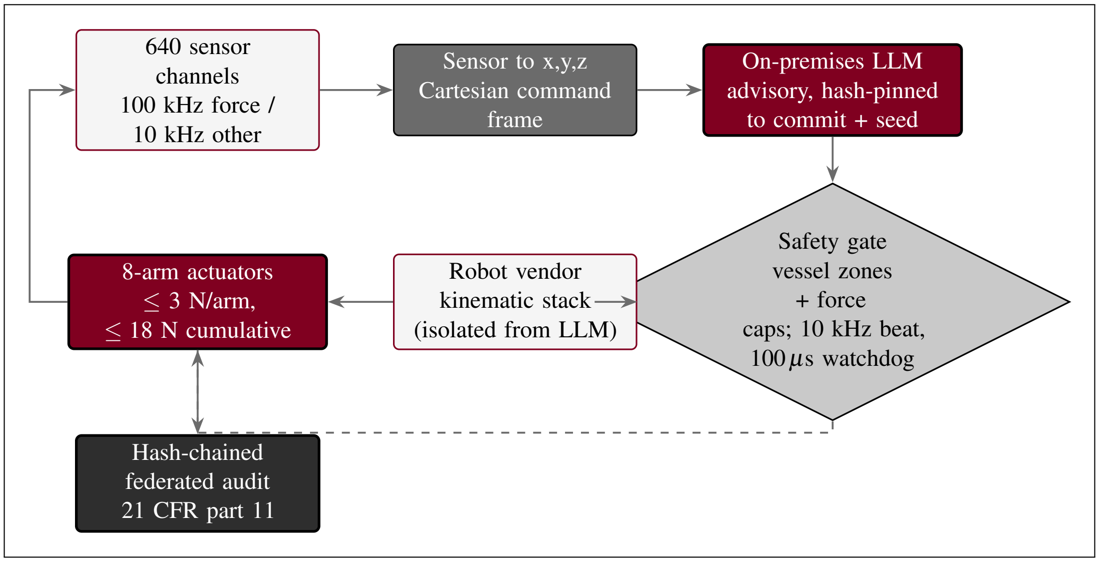

Figure 11. On-premises LLM advisory control loop. Sensor data is mapped to a Cartesian command frame; the LLM issues hash-pinned advisory commands from outside the vendor kinematic stack (second-opinion oracle isolation); and a safety gate enforces the vessel zones, the force caps, the 10 kHz heartbeat, and the 100 μs watchdog before any actuator moves. Actuator telemetry returns to the sensor stack, and every step is written to a hash-chained federated audit trail under 21 CFR part 11 [5].

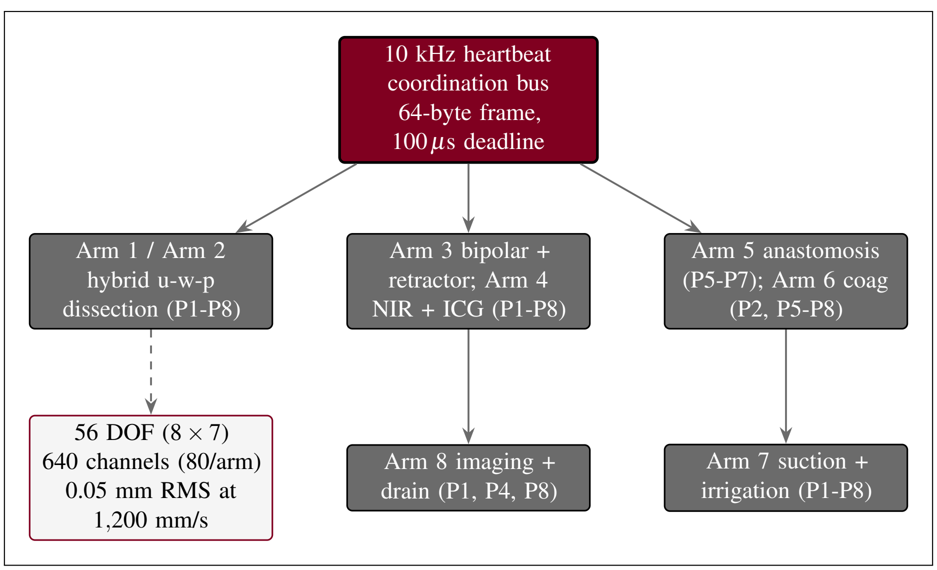

Figure 12. Eight-arm platform architecture. Eight cooperating arms (56 degrees of freedom, 640 sensor channels at 80 per arm, 0.05 mm root mean square positioning at 1,200 mm/s) are coordinated by a 10 kHz heartbeat bus; each arm's tool assignment and operative-phase coverage drive the Schedule of Activities.

The per-arm tool assignment crossed with the eight operative phases (P1 through P8) is given in Table 5, and ten representative channels drawn from the 640-channel inventory are given in Table 6.

Table 5: Per-arm tool assignment and operative-phase coverage for the eight-arm LLM-directed Whipple platform (PancreSpeed II).

Arm	**Primary tool**	**Phase coverage**	**Sensor channels (10 of 80)**
Arm 1	Hybrid u-w-p end-effector	P1 to P8 (active dissect)	Tip force, joint torque, ID
Arm 2	Hybrid u-w-p end-effector	P1 to P8 (active dissect)	Tip force, joint torque, ID
Arm 3	Bipolar + retractor	P1 to P8 (vessel control)	Tip force, current, force
Arm 4	NIR + ICG imaging probe	P1 to P8 (imaging)	ICG fluorescence, NIR depth
Arm 5	Anastomosis probe	P5, P6, P7 (idle others)	Ring tension, ring strain
Arm 6	Bipolar coag + cautery	P2, P5, P6, P7, P8	Tip force, RF power
Arm 7	Suction + irrigation	P1 to P8	Suction pressure, pH
Arm 8	Final images + drain placement	P1, P4, P8 (light load)	ICG fluorescence, drain ID

Table 6: Ten representative sensor channels drawn from the 640-channel inventory (80 channels per arm).

Arm	**Channel name**	**Rate**	**Unit**	**Dynamic range**
1	tip_force_x	100 kHz	N	0 to 3.0 N
1	joint_torque_3	10 kHz	N·m	−50 to 50
2	tip_force_z	100 kHz	N	0 to 3.0 N
3	retractor_force	10 kHz	N	0 to 5.0 N
4	icg_fluorescence	10 kHz	a.u.	0 to 65,535
4	nir_depth	10 kHz	mm	0 to 50
5	ring_tension	10 kHz	N	0 to 1.0 N
6	rf_power	10 kHz	W	0 to 80
7	suction_pressure	10 kHz	kPa	−90 to 0
8	drain_id	10 kHz	enum	0 to 4

Physical AI System Description fields (21 CFR §312.23(g)). Because the device participates in the investigational drug delivery pathway, the combined IND/IDE carries a full Physical AI System Description meeting the eleven fields of 21 CFR §312.23(g) [6]. The fields, and the location of the corresponding evidence in this submission, are summarized below; the multicenter conduct adds fleet harmonization and a federated audit across the eight centers.

(i) *System identification and classification:* manufacturer, model designation, software version, surgical robot category, and the current Unified Safety Level (USL) score computed across simulation switching, AI integration, cross-robot sharing, and clinical-trial collaboration.

(ii) *Role in drug administration:* secondary role (the device does not itself dispense daraxonrasib; it conducts the resection and reconstruction and generates the operative and pathology inputs to the restart advisory), with permitted autonomy levels and human-oversight requirements for each procedure.

(iii) *Hardware specifications:* 56 degrees of freedom, 0.05 mm root mean square positioning at 1,200 mm/s, 100 kHz force sensing, the $\leq$ 3 N per-arm and $\leq$ 18 N cumulative tip-force limits, and end-effector details.

(iv) *Software and AI/ML specifications:* the operating and middleware stack, the control algorithms, and the on-premises advisory LLM together with its training data characteristics, validation methodology, performance metrics, and Predetermined Change Control Plan (PCCP).

(v) *Simulation validation data:* the upgraded Phase 0 campaign results, including at least 5000 simulated procedures reproduced across at least three independent frameworks and the tightened sim-to-real gap quantification (trajectory gap $<$ 1.5 mm, tip-force gap $<$ 0.4 N).

(vi) *Digital twin specifications:* the digital-twin methodology, calibration sources, update frequency, and validation metrics used for operative rehearsal and intra-operative guidance.

(vii) *Safety systems:* the emergency-stop mechanisms (hardware and software), the force and torque limits, the five-vessel no-fly gate, collision detection, workspace-boundary enforcement, fail-safe behaviors, and the pre-procedure safety matrix.

(viii) *Cybersecurity assessment:* the network architecture, deny-by-default authentication and authorization, encryption, vulnerability management, and incident response.

(ix) *MCP integration:* the Model Context Protocol server topology, conformance level, robot capability profile, and the hash-chained federated audit-trail implementation under 21 CFR part 11 [5].

(x) *Human-robot interaction protocols:* the operator control stations, status displays, alarm systems, handoff procedures, and the training requirements for operating and supervising personnel across the fleet.

(xi) *Maintenance and calibration:* the scheduled maintenance intervals, calibration procedures, out-of-service criteria, per-site qualification, and re-qualification requirements (§6.2).

7.1.2 Dosing and Administration

Daraxonrasib dosing. In Arm A daraxonrasib is administered orally once daily at the fixed RP2D of 300 mg. Because the RP2D is already established in the Phase 1 predicate and anchored to the RASolute 302 pan-KRAS program [1, 14], there is no 3+3 dose escalation in this Phase 2 study; every participant in Arm A receives the same fixed daily dose, and the trial tests comparative efficacy rather than dose-finding.

Perioperative pause-and-restart advisory. In Arm A daraxonrasib is paused before surgery and restarted postoperatively on a day chosen by an advisory that is LLM-bound and human-confirmed, never autonomous. The advisory takes two inputs: the pancreaticojejunostomy ISGPS postoperative pancreatic-fistula grade (A, B, or C) [11] and the serum daraxonrasib trough relative to a 0.5 ng/mL threshold. It recommends a restart on postoperative day T+7, T+14, or T+21 (or a continued hold): T+7 for a Grade A fistula with a sub-threshold trough, T+14 for a Grade B fistula or a delayed serum rise, and T+21 or continued hold for a Grade C fistula, which also maps to the drug stop rule of §7. The advisory logic is shown in Figure 13.

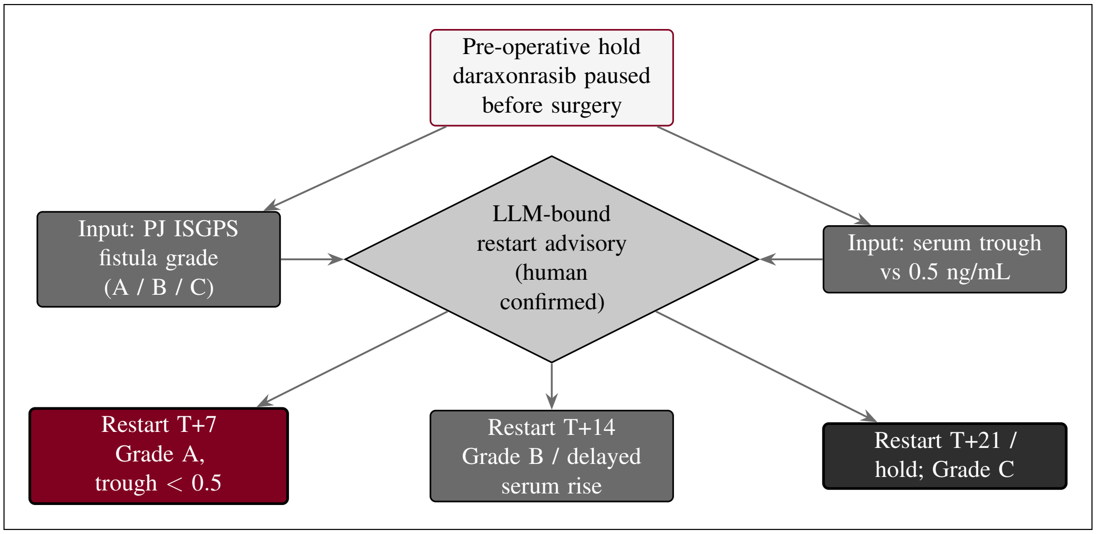

Figure 13. Daraxonrasib perioperative pause-and-restart advisory in Arm A. The restart day is keyed to the pancreaticojejunostomy ISGPS fistula grade and the serum trough relative to 0.5 ng/mL: T+7 for Grade A with a sub-threshold trough, T+14 for Grade B or a delayed serum rise, and T+21 or continued hold for Grade C. The advisory is bound and human-checked, not autonomous.

Device administration: operative phases and the three anastomoses. The device has no pharmacologic dose; its administration is the conduct of the pancreaticoduodenectomy across eight operative phases (P1 through P8) under continuous human oversight. Phases P1 through P4 comprise the dissection and specimen removal (Arms 1 and 2 dissecting under Arm 3 vessel control and Arm 4 near-infrared indocyanine-green imaging); the specimen is removed at the end of P4 and reconstruction begins. The three reconstructive anastomoses are then constructed in sequence under closed-loop ring-tension control by the Arm 5 anastomosis probe, which issues a soft warning whenever measured ring tension falls outside the controller band: the duct-to-mucosa pancreaticojejunostomy (PJ) in P5 with a band of 0.30 to 0.60 N, the end-to-side hepaticojejunostomy (HJ) in P6 with a band of 0.20 to 0.50 N, and the antecolic gastrojejunostomy (GJ) in P7 with a band of 0.40 to 0.80 N; P8 covers final imaging, hemostasis, and drain placement. The pancreaticojejunostomy is the dominant determinant of postoperative pancreatic-fistula risk and is therefore the anastomosis on which the fistula-grade input to the drug advisory is based. The reconstruction sequence and the ring-tension bands are shown in Figure 14.

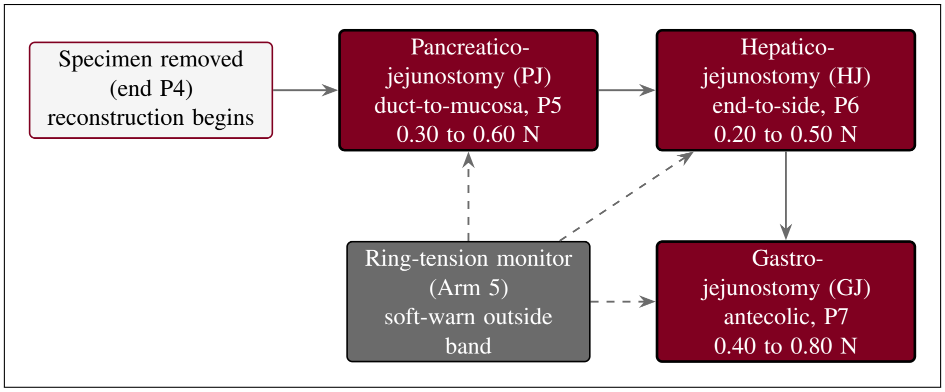

Figure 14. Three reconstructive anastomoses and their controller ring-tension bands: the duct-to-mucosa pancreaticojejunostomy (0.30 to 0.60 N), the end-to-side hepaticojejunostomy (0.20 to 0.50 N), and the antecolic gastrojejunostomy (0.40 to 0.80 N), each guarded by the Arm 5 ring-tension monitor issuing a soft warning outside the band.

7.2 Preparation, Handling, Storage, and Accountability

7.2.1 Acquisition and Accountability

Daraxonrasib is supplied by the sponsor to the investigational-product pharmacy at each participating site and is dispensed only by the site pharmacist or an authorized designee against a written prescription tied to an enrolled, Arm A participant. The pharmacy maintains a drug-accountability log recording, for every unit, the lot number, the quantity received, the quantity dispensed, the quantity returned, and the quantity destroyed, reconciled against shipment records at each monitoring visit. Returned and unused product is retained for reconciliation and destroyed only after sponsor authorization. The modified FOLFIRINOX components of Arm B are obtained and accounted for through standard institutional oncology pharmacy practice. The investigational device is provided under the IDE and is not a consumable; its accountability is established through the asset record, the per-procedure pre-procedure safety matrix, and the hash-chained federated audit trail that captures every operative use across the fleet (§6.4). Custody of all components is restricted to authorized, trained personnel, and all transfers are documented.

7.2.2 Formulation, Appearance, Packaging, and Labeling

Daraxonrasib is provided as oral tablets in the strengths required to compose the 300 mg once-daily RP2D. Tablets are packaged in child-resistant, light-protective containers labeled in accordance with 21 CFR §312.6, bearing the caution statement “Caution: New Drug. Limited by Federal (or United States) law to investigational use,” together with the protocol number, lot number, storage conditions, and a unique container identifier. Because this is an open-label study, the participant-facing label identifies the product and the assigned daily dose; no label information is used to influence the masked outcome assessors described in §6.5. The investigational device and its single-use accessories are labeled per 21 CFR part 812 with the investigational-device caution statement and are not relabeled or repackaged at the site.

7.2.3 Product Storage and Stability

Daraxonrasib is stored at controlled room temperature in a secured, access-controlled, temperature-monitored location within the investigational-product pharmacy, in its original light-protective packaging, until dispensed. Continuous temperature monitoring with documented excursion handling is maintained; any excursion outside the labeled range is quarantined and is not dispensed until the sponsor confirms continued suitability against the stability data on file. Stability is supported by the sponsor's stability program, and product is dispensed only within its labeled expiry or retest date. The investigational device is stored and maintained in its qualified clinical environment between procedures under the maintenance and calibration program below, with the same qualification applied identically at every site in the harmonized fleet.

7.2.4 Preparation and Device Maintenance and Calibration

Drug preparation. No compounding or reconstitution is required for daraxonrasib; preparation consists of the pharmacist verifying the participant identity, the RP2D, and the expiry, then dispensing the once-daily oral dose with administration instructions. Tablets are swallowed whole with water and are not crushed, split, or chewed unless a future protocol amendment provides validated instructions. Modified FOLFIRINOX is prepared per institutional oncology pharmacy standards.

Device maintenance and calibration (21 CFR §312.23(g)(xi)). Before each clinical procedure the system must pass the pre-procedure safety matrix: verification of the 640-channel sensor stack, confirmation of the per-arm force calibration against the ≤ 3 N per-arm and ≤ 18 N cumulative caps, exercise of the 10 kHz heartbeat with its 100 μs watchdog, confirmation of the five-vessel no-fly gate, a successful software and hardware E-stop self-test, and confirmation that the positioning accuracy remains within the 0.05 mm root mean square specification at 1,200 mm/s. Scheduled maintenance, periodic force and positional calibration, per-site qualification, and fleet harmonization verification are performed at the manufacturer-specified intervals and documented. A system that fails any pre-procedure check, exhibits an out-of-tolerance calibration, or experiences a hard-stop E-stop event is taken out of service and is returned to clinical use only after re-qualification against the same matrix and sponsor authorization. All maintenance, calibration, and out-of-service events are recorded in the hash-chained federated audit trail.

7.3 Measures to Minimize Bias: Randomization and Blinding

Allocation uses central permuted-block randomization 1:1, stratified by resectability (resectable versus borderline-resectable), KRAS allele (G12D versus other G12), neoadjuvant therapy (yes versus no), and site, so that the central mechanism prevents allocation bias and balances the strata across arms [27]. Treatment is open-label because a surgical-system intervention cannot be masked from the operating team or the participant; bias control is therefore concentrated downstream of the intervention. The imaging-based endpoints pass through blinded independent central review (BICR); the R0 resection status and the major pathologic response (MPR) are determined on central masked pathology by reviewers who are masked to the assigned arm, to the intra-operative device telemetry, and to the LLM advisory record, with discordant or indeterminate margins adjudicated by an independent masked second pathologist; and all confirmatory endpoints pass through blinded endpoint adjudication independent of the operative team. In this way the open-label delivery does not propagate into outcome ascertainment.

7.4 Study Intervention Compliance

Daraxonrasib adherence. In Arm A, adherence is documented through the dispensing and accountability log, returned-tablet counts at each visit, and a participant dosing diary reconciled against the pharmacy record. Adherence is corroborated objectively by the serum daraxonrasib assay, which both confirms exposure and supplies the trough value used by the restart advisory (§6.1.2); a documented trough discordant with the reported intake prompts adherence counseling and is recorded in the case report form. Adherence to modified FOLFIRINOX in Arm B is captured through standard cycle administration records. Material non-adherence in either arm is captured as a protocol deviation and informs the per-protocol analysis (§9).

Device compliance. Device compliance is enforced before and during every procedure rather than reconstructed afterward. No operative phase may begin until the pre-procedure safety matrix passes in full, and the 10 kHz heartbeat, the 100 μs watchdog, the force caps, and the five-vessel no-fly gate operate continuously throughout the case. Every advisory command, every safety-gate verdict, every force and trajectory sample at the recorded rates, and every E-stop event is written to a 21 CFR part 11 hash-chained, append-only federated audit trail [5], so that compliance with the cyber-physical envelope is verifiable for each case across the fleet; any deviation is detectable and time-stamped. The audit trail also satisfies the preservation requirement that applies around a reportable Physical AI adverse event (§8).

7.5 Concomitant Therapy

All medications, biologics, blood products, radiotherapy, and clinically significant non-pharmacologic interventions taken from the time of informed consent through the end of safety follow-up are recorded in the case report form with indication, dose, route, and dates. Standard perioperative supportive care is permitted and encouraged in both arms, including antiemetics, analgesia, venous thromboembolism prophylaxis, peri-operative antimicrobial prophylaxis, proton-pump inhibitors, and pancreatic-enzyme replacement as clinically indicated, with attention in Arm A to agents that may interact with daraxonrasib exposure. Prohibited during study participation are systemic anticancer therapy outside the assigned regimen (for Arm A, cytotoxic chemotherapy, other targeted agents, or immunotherapy beyond the protocol; for Arm B, agents beyond the modified FOLFIRINOX standard), other investigational agents or investigational devices, and any concomitant use of a second surgical robotic system for the index operation. Strong inducers or inhibitors of the daraxonrasib principal metabolic pathway are avoided where feasible in Arm A; when an interacting agent is clinically unavoidable, its use is documented and reviewed against exposure data.

7.5.1 Rescue Medicine

Rescue is available along two axes. For surgical and device-related complications, the operative team may at any time convert to manual or open technique (§7) and deliver standard rescue care: transfusion and hemostatic resuscitation for bleeding, vasoactive support for hemodynamic instability, percutaneous or operative drainage and broad-spectrum antimicrobials for a clinically relevant postoperative pancreatic fistula or intra-abdominal collection, and somatostatin-analogue therapy where indicated for high-output fistula. For drug toxicity, daraxonrasib in Arm A is held or discontinued per the stopping rules (§7) and managed symptomatically with the appropriate supportive agents (for example antidiarrheals and aggressive dermatologic care for the rash and gastrointestinal toxicities characteristic of the class), and modified FOLFIRINOX in Arm B is dose-modified or held per its established toxicity management, with dose interruption and reduction governed by the protocol. All rescue interventions are recorded as concomitant therapy and, where the criteria are met, as adverse or serious adverse events.

8 Intervention and Participant Discontinuation/Withdrawal

This section defines the prospectively specified rules for discontinuing the study intervention (the device, the drug, or both), the procedures governing a participant's withdrawal from the study, and the handling of participants lost to follow-up. Discontinuation of an intervention is distinct from withdrawal from the study: a participant who stops one or both components of the assigned regimen for a protocol-defined reason remains in the study, and remains in the randomized intention-to-treat (ITT) population, for safety and survival follow-up unless consent for that follow-up is withdrawn. Because this is a randomized, controlled trial, the rules below apply within the arm to which the participant was randomized; the device stopping rules apply to Arm A, and the drug stopping rules are specified separately for daraxonrasib in Arm A and for modified FOLFIRINOX in Arm B.

8.1 Discontinuation of Study Intervention

Device stopping rules. For a participant randomized to Arm A, the investigational device is discontinued for the index procedure, with conversion to the appropriate fallback technique, whenever any of the following prespecified conditions occurs. These rules implement the Physical AI stopping criteria of 21 CFR §312.23(g)(i)(v) and the divergence-reporting threshold of §312.32(g) [6], and the federated hash-chained audit trail is preserved across each event so that the behavior can be reconstructed across the eight-site fleet.

- *Conversion to open or manual technique.* Any intra-operative judgment by the supervising surgeon that patient safety or oncologic adequacy is better served by manual robotic or open conversion (including uncontrolled bleeding, unfavorable anatomy, or loss of a critical view) ends device-directed operation immediately; conversion is always available and is never itself a protocol violation.
- *Repeated force-limit excursions.* Repeated excursions against the ≤ 3 N per-arm or ≤ 18 N cumulative cross-arm tip-force caps, or any sustained or consecutive pattern of force-clamp activations indicating that the case cannot proceed within the force envelope, trigger discontinuation.
- *No-fly-zone violations.* Any breach of a vascular no-fly zone around a named vessel (superior mesenteric vein or portal vein at 1.0 mm; hepatic artery, celiac axis, or superior mesenteric artery at 1.5 mm), or a recurring pattern of no-fly approaches, ends device-directed dissection in that field.
- *Sim-to-real divergence beyond twice tolerance.* Any instance in which the observed device behavior diverges from the validated simulation prediction by more than twice the sim-to-real gap tolerance (a trajectory deviation greater than 3 mm or a force discrepancy greater than 0.8 N) triggers discontinuation and a report under §312.32(g) [6].
- *Hard-stop E-stop event.* Any hard-stop cross-arm E-stop (the ≤ 3 ms full-platform halt, whether software- or hardware-initiated) ends the device-directed segment; resumption with the device requires that the cause be identified and that the system re-pass the pre-procedure safety matrix (§6.2), failing which the case is completed by conversion.

Following any device discontinuation, the operation is completed by the most appropriate conventional means, the federated hash-chained audit trail is preserved across the event window (§10.10), and the event is reviewed by the Physical AI Safety Review Committee and the data and safety monitoring board (DSMB) before the device is used in a subsequent case at any site. A participant whose index procedure is converted is analyzed in Arm A as randomized (§9).

Drug stopping rules. For a participant in Arm A, daraxonrasib is discontinued for an unacceptable toxicity that does not resolve with protocol dose interruption and reduction, for a confirmed International Study Group of Pancreatic Surgery (ISGPS) Grade C postoperative pancreatic fistula [11] (which precludes restart), for pregnancy, for intercurrent illness that contraindicates continuation, or at the participant's request. For a participant in Arm B, modified FOLFIRINOX is managed, interrupted, dose-reduced, or discontinued according to standard toxicity management for that regimen [15]. In either arm, dose interruptions and reductions for lower-grade toxicity follow the applicable algorithm and do not by themselves mandate permanent discontinuation. A participant who permanently discontinues the assigned drug, in either arm, remains in the study and in the ITT population for survival follow-up to the 24-month endpoint.

8.2 Participant Discontinuation/Withdrawal from the Study

A participant may withdraw from the study at any time, for any reason, without penalty and without loss of any benefit to which the participant is otherwise entitled, and a participant may separately withdraw consent for continued study-specific procedures while permitting ongoing survival follow-up through the medical record. The investigator may withdraw a participant for safety, for intercurrent illness, for significant non-compliance that compromises participant safety or data integrity, for loss of eligibility, or at the sponsor's direction if the study is terminated. The reason for and date of any discontinuation or withdrawal are recorded in the case report form (CRF), and a participant who withdraws is asked to complete a final safety assessment where feasible.

Data and specimens collected up to the point of withdrawal are retained and analyzed in accordance with the consent and applicable regulation; withdrawal of consent for future participation does not require deletion of data already collected. The relationship of discontinuation and withdrawal to the analysis populations (the ITT, modified ITT, per-protocol, and safety populations) is defined in the statistical considerations (§9). Because this is a randomized ITT trial, randomized participants are analyzed in the arm to which they were randomized, regardless of the intervention actually received or completed, and are not replaced; no participant who discontinues an intervention or withdraws is substituted by a newly randomized participant.

8.3 Lost to Follow-Up

A participant is considered lost to follow-up only after the site has made and documented at least three contact attempts by telephone across separate days, followed by a written contact attempt to the last known address, over the 24-month overall-survival window. Each attempt is recorded with its date and outcome in the CRF. Before declaring a participant lost to follow-up, the site attempts, with appropriate authorization, to ascertain vital status from the medical record, the referring provider, and permissible public sources, because overall survival is a key endpoint of the study. A participant who cannot be reached after these attempts is classified as lost to follow-up, with the last date known alive recorded. The handling of the resulting missing data, including the censoring rules for progression-free and overall survival and the sensitivity analyses used to assess the impact of missingness, is specified in §9.

9 Study Assessments and Procedures

This section specifies the efficacy and safety assessments, the device-telemetry measurements unique to the Physical AI component, and the adverse-event and unanticipated-problem machinery for the study. The timing of every assessment is governed by the Schedule of Activities (§1.3); this section defines what is measured, how it is graded, and how it is reported. Because the study is conducted across eight academic high-volume hepatopancreatobiliary centers, every endpoint that admits central reading is read centrally: a central imaging core performs the blinded independent central review (BICR), a central pathology core performs the masked synoptic margin and regression reads, and central laboratories perform the biomarker and clinical-chemistry assays, so that ascertainment is uniform across sites and independent of the open-label assignment.

9.1 Efficacy Assessments

The primary efficacy endpoint is progression-free survival (PFS), the time from randomization to the first documented radiographic disease progression by RECIST version 1.1 or death from any cause, whichever occurs first. Imaging is acquired on contrast-enhanced cross-sectional studies at the protocol-specified timepoints and is read by the central imaging core; the investigator-assessed read is the primary analysis and the BICR read supports a prespecified sensitivity analysis (§9.4.2). Overall survival (OS), the time from randomization to death from any cause, is followed to the 24-month endpoint and is a key secondary endpoint, with the 2025 Dutch nationwide robotic pancreaticoduodenectomy cohort retained as an external descriptive reference [10].

The surgical and biomarker efficacy endpoints are assessed centrally. The margin-negative (R0) resection rate is determined by central masked synoptic pathology, with the pathologist masked to the assigned arm and to the device telemetry and with masked independent adjudication of discordant or indeterminate margins. The major pathologic response (MPR) is graded by the same central pathology core on the resection specimen. Postoperative pancreatic fistula is graded by the International Study Group of Pancreatic Surgery (ISGPS) definition as a biochemical leak (no longer classified as a clinically relevant fistula), Grade B (requiring a change in management), or Grade C (requiring reoperation or causing organ failure or death) [11]; the clinically relevant (Grade B or C) rate is a key secondary endpoint. Circulating tumor DNA (ctDNA) bearing the participant's KRAS mutation is measured by a central laboratory, and week-12 molecular clearance, the conversion from a detectable to an undetectable KRAS ctDNA level by week 12, is a key secondary endpoint [12].

Anastomosis quality is a prespecified Arm A device-quality assessment and is recorded for each of the three reconstructions against its controller ring-tension band: the duct-to-mucosa pancreaticojejunostomy (0.30 to 0.60 N), the end-to-side hepaticojejunostomy (0.20 to 0.50 N), and the antecolic gastrojejunostomy (0.40 to 0.80 N), as constructed under continuous Arm A telemetry (§6). The pancreaticojejunostomy ring tension is the dominant fistula determinant and is recorded against the fistula and biomarker reads above. Health-related quality of life is measured with the EORTC QLQ-C30 core questionnaire and the PAN26 pancreatic-cancer module at the protocol-specified timepoints [16].

9.2 Safety and Other Assessments

Clinical safety assessments comprise vital signs, a directed physical examination, hematology, serum chemistry including hepatic and pancreatic indices, coagulation studies, the serum daraxonrasib trough assay for Arm A, and protocol-specified imaging, at the frequencies in the Schedule of Activities, with clinical-chemistry and biomarker assays routed to the central laboratories. In addition to these conventional assessments, the Arm A device generates a continuous safety telemetry stream that is a source of protocol observations under 21 CFR §312.23(g): per-arm and cumulative force profiles against the ≤ 3 N per-arm and ≤ 18 N cumulative cross-arm caps, trajectory accuracy against specification, every vascular safety-zone verdict, every heartbeat and watchdog event, and every E-stop activation, each captured at its recorded rate in the hash-chained audit trail. Pre-procedure safety-matrix results are recorded for each Arm A case.

The vascular safety-zone gate ticks at 10 kHz and evaluates the instrument-tip proximity to each of five named vessels against three concentric thresholds, with the verdict driving the response: a soft-warning zone at 3.0 mm raises an LLM advisory, a no-fly zone (1.0 mm at the superior mesenteric and portal veins; 1.5 mm at the hepatic artery, celiac axis, and superior mesenteric artery) blocks the command, and a hard-stop zone at 5.0 mm forces a ≤ 3 ms cross-arm E-stop. The gate exercises all five vessels across all eight phases of the operation, and every verdict is written to the audit trail. The gate is shown in Figure 15.

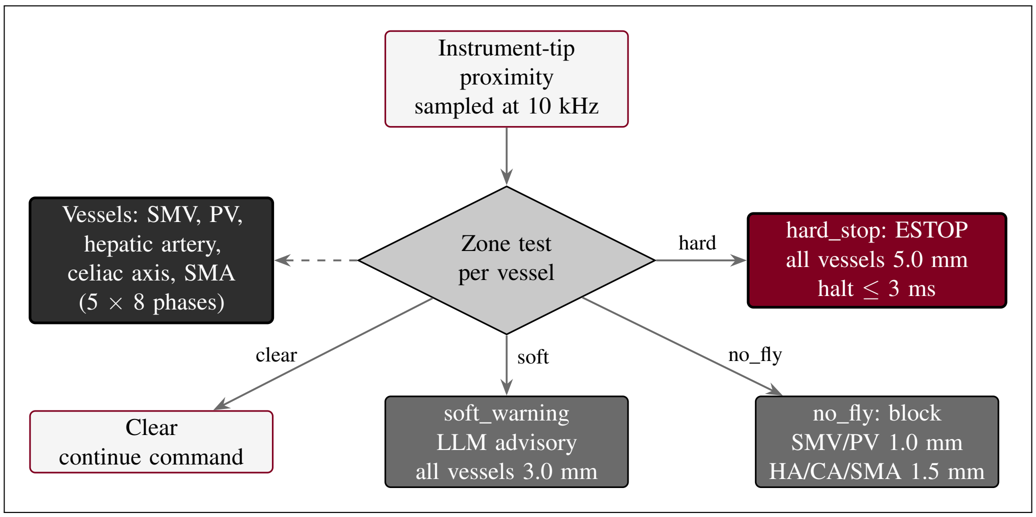

Figure 15. Five-vessel vascular safety-zone gate for Arm A. Three concentric proximity zones around each named vessel drive the response: a soft-warning zone (3.0 mm) raises an LLM advisory, a no-fly zone (1.0 mm at the superior mesenteric and portal veins; 1.5 mm at the hepatic artery, celiac axis, and superior mesenteric artery) blocks the command, and a hard-stop zone (5.0 mm) forces a ≤ 3 ms cross-arm E-stop. The gate is sampled at 10 kHz across all eight phases of the operation.

The halt chain that backs the gate is layered. The 10 kHz heartbeat bus broadcasts a 64-byte frame on a 100 μs per-arm watchdog deadline; a frame that is on time continues the command state, whereas a missed or out-of-parameter frame parks the affected arm within 50 μs and escalates to a cross-arm E-stop that halts the platform in ≤ 3 ms at a 0.0 N force clamp. A software-independent hardware E-stop meeting the ≤ 500 ms requirement backs the software chain. The architecture is shown in Figure 16.

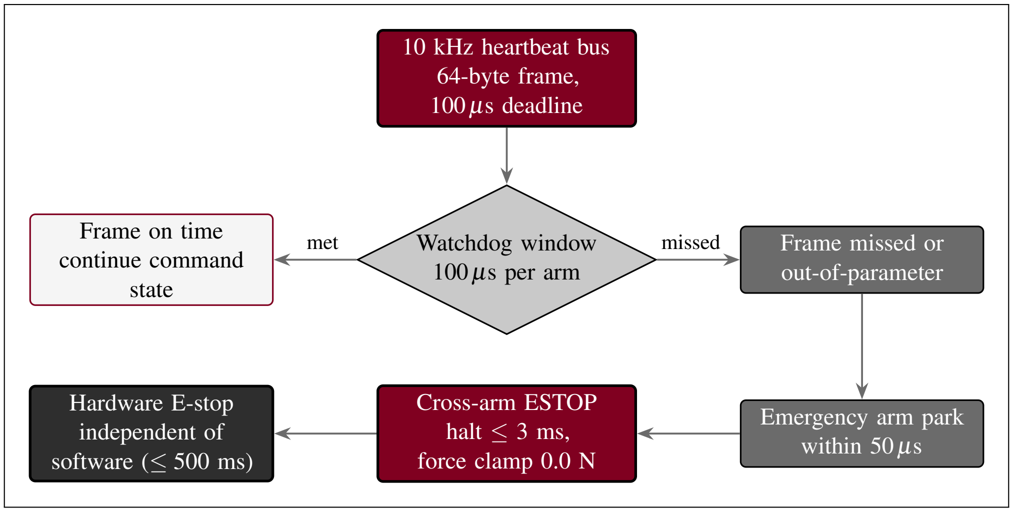

Figure 16. Heartbeat, watchdog, and E-stop architecture for Arm A. A 10 kHz heartbeat bus with a 100 μs per arm watchdog continues the command state on an on-time frame; a missed frame parks the arm within 50 μs and escalates to a cross-arm E-stop that halts the platform in ≤ 3 ms at a 0.0 N force clamp, backed by a software-independent hardware E-stop meeting the ≤ 500 ms requirement.

9.3 Adverse Events and Serious Adverse Events

Adverse events are captured through two parallel and equally binding streams: a conventional pharmacovigilance stream for clinical events in both arms, and a Physical AI safety stream for device- and AI-related events in Arm A. The combined workflow is shown in Figure 17.

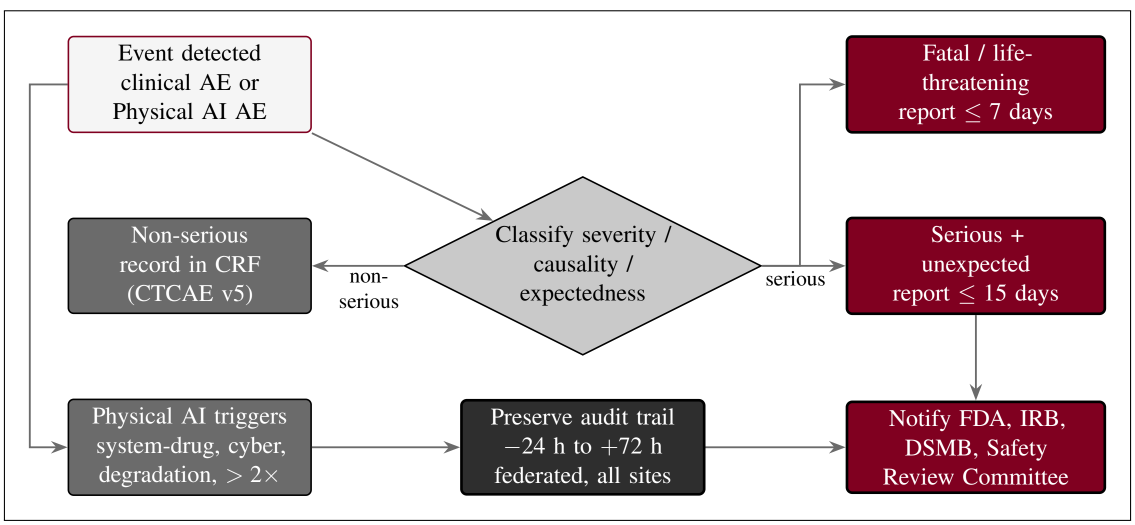

Figure 17. Adverse-event and Physical AI AE reporting workflow. Clinical events are graded by CTCAE v5 and reported on the 15-day (serious, unexpected) or 7-day (fatal, life-threatening) IND/IDE timelines, while Physical AI triggers add their own reports, each preserving the hash-chained audit trail from 24 hours before to 72 hours after the event, federated across the eight sites, with notification to the FDA, the IRB, the DSMB, and the Physical AI Safety Review Committee.

9.3.1 *Definition of Adverse Events (AE)*

An adverse event is any untoward medical occurrence in a participant administered the study intervention, whether or not considered related to it; the term encompasses any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the intervention. Clinical adverse events in both arms are graded for severity by the Common Terminology Criteria for Adverse Events (CTCAE) version 5.0. In addition, a Physical AI adverse event, per 21 CFR §312.32(f), is any untoward occurrence associated with the operation of the Arm A device during the investigation, whether or not caused by it, including robotic malfunction, AI prediction error, sensor failure, communication loss, unauthorized system access, digital-twin divergence, and any event requiring activation of the emergency stop [6].

9.3.2 Definition of Serious Adverse Events (SAE)

A serious adverse event is any adverse event that results in death, is life-threatening, requires inpatient hospitalization or prolongation of existing hospitalization, results in persistent or significant disability or incapacity, is a congenital anomaly or birth defect, or is an important medical event that may jeopardize the participant and may require intervention to prevent one of the foregoing outcomes. A serious Physical AI adverse event, per §312.32(f), is a Physical AI adverse event that results in, or has the potential to result in, any serious-adverse-event outcome, or that results in an unplanned interruption of an investigational drug-administration procedure requiring clinical intervention, an uncontrolled release of the investigational drug, or compromise of the device's sterile field during a surgical procedure.

9.3.3 Classification of an Adverse Event

Each adverse event is classified along four axes. Severity is graded by CTCAE version 5.0 (Grade 1 mild through Grade 5 fatal). Causality is assessed separately in relation to the drug and in relation to the device, each as related or unrelated (with relatedness denoting a reasonable possibility that the component caused the event), so that a single event may carry independent drug and device causality attributions; in Arm B, which has no investigational device, causality is assessed in relation to the drug and to the procedure. Expectedness is judged against the reference safety information (the daraxonrasib investigator's brochure for the drug and the Physical AI supplement and System Description for the device), and an event not listed there at the observed specificity or severity is classified as unexpected. Seriousness is determined against the criteria in §8.3.2. The combination of unexpected, serious, and at least possibly related defines a suspected adverse reaction reportable on the expedited timelines below.

9.3.4 Time Period and Frequency for Assessment and Follow-Up

Adverse events are solicited and recorded from the time of informed consent and are assessed at every study contact. The primary intensive safety-collection window spans from consent through the 30-day postoperative safety visit. Beyond the 30-day window, serious adverse events, events of special interest, Physical AI adverse events, and new anticancer therapy continue to be collected through the long-term follow-up period to the 24-month overall-survival endpoint. Each event is followed until resolution, stabilization, return to baseline, or a determination that no further change is expected, or until the participant is lost to follow-up or withdraws consent.

9.3.5 Adverse Event Reporting

All adverse events, serious and non-serious, are recorded in the case report form with onset and resolution dates, severity grade, drug and device causality, expectedness, seriousness, action taken with each intervention, and outcome. Routine (non-serious) events are captured on the adverse-event case report form at the scheduled visits and do not require expedited transmission. Device-telemetry-derived observations in Arm A that meet the adverse-event definition are entered with reference to the corresponding audit-trail records.

9.3.6 Serious Adverse Event Reporting

Serious adverse events are reported by the investigator to the sponsor within 24 hours of awareness. The sponsor submits IND/IDE safety reports to the FDA on the regulatory timelines: any unexpected fatal or life-threatening suspected adverse reaction is reported as soon as possible but no later than 7 calendar days after initial receipt of the information, and any other serious and unexpected suspected adverse reaction is reported no later than 15 calendar days after the determination that the event is reportable. In addition to the clinical SAE pathway, six Physical AI reporting triggers are reported under §312.32(g): (i) any serious Physical AI adverse event, on the 7- or 15-day timeline as applicable; (ii) any Physical AI system-drug interaction event causing incorrect drug administration (wrong dose, route, timing, or patient), regardless of seriousness; (iii) any cybersecurity incident, within 15 calendar days (or 7 days if it has the potential to cause patient harm); (iv) sustained AI model performance degradation below the pre-specified acceptance criteria for three or more consecutive procedures or a cumulative 24 hours; (v) any sim-to-real divergence beyond twice the validated gap tolerance (trajectory deviation > 3 mm or force discrepancy > 0.8 N); and (vi) any digital-twin discrepancy beyond the pre-specified accuracy thresholds. For every Physical AI adverse event reported, the complete hash-chained audit trail covering the period from 24 hours before the event to 72 hours after the event is preserved under 21 CFR part 11, federated across the eight sites, and made available to the FDA on request [5].

9.3.7 Events of Special Interest

The following events are designated events of special interest and are tracked and reported with the same diligence as serious adverse events regardless of whether they independently meet seriousness criteria: vascular injury (including any hard-stop E-stop precipitated by a vascular safety-zone breach), clinically relevant postoperative pancreatic fistula (ISGPS Grade B or C) [11], conversion from device-directed to manual or open technique, any cybersecurity incident affecting the device, and any model-performance degradation. Each is reviewed by the Physical AI Safety Review Committee, which convenes at intervals not exceeding 90 days during active enrollment, and is reported to the data safety monitoring board (DSMB).

9.3.8 Reporting of Pregnancy

Pregnancy is not an adverse event but is reported because of the potential exposure of an embryo or fetus to the study chemotherapy. Participants of childbearing potential and partners of participants must use highly effective contraception during treatment and for the protocol-specified interval afterward. Any pregnancy in a participant or a participant's partner occurring during treatment or within that interval is reported by the investigator to the sponsor within 24 hours of awareness and is followed to outcome; a pregnancy with an associated serious adverse event, or an abnormal pregnancy outcome, is reported on the applicable expedited SAE timeline.

9.4 Unanticipated Problems

9.4.1 Definition

An unanticipated problem is any incident, experience, or outcome that is unexpected (not described in the protocol, consent, investigator's brochure, or Physical AI System Description in terms of nature, severity, or frequency given the procedures and population), related or possibly related to participation, and that suggests the research places participants or others at a greater risk of harm (including physical, psychological, economic, or social harm) than was previously known or recognized. For the device, the Physical AI hold grounds (repeated uncorrected safety failures, a confirmed cybersecurity compromise of drug-administration parameters or system integrity, and a simulation-validation deficiency that undermines the pre-clinical safety basis) are treated as paradigm unanticipated problems.

9.4.2 Reporting

Unanticipated problems are reported by the investigator to the reviewing IRB and, as applicable, to the sponsor and the FDA in accordance with the institutional policy and the IND/IDE requirements. An unanticipated problem that is also a serious and unexpected suspected adverse reaction is reported on the 7- or 15-day timeline of §8.3.6; other unanticipated problems are reported promptly and ordinarily within the institution's defined window (typically no later than the next scheduled continuing review, and sooner when the problem indicates an immediate hazard). Reports that involve the device preserve the hash-chained audit trail across the event window, federated across the eight sites [5]. The Physical

AI Safety Review Committee and the DSMB review each unanticipated problem and recommend any change to device operation, human-oversight requirements, Unified Safety Level thresholds, or the protocol.

9.4.3 Reporting to Participants

When an unanticipated problem affects the rights, safety, or welfare of enrolled participants, or could affect a participant's willingness to continue, the information is communicated to affected participants in a timely manner, ordinarily through an IRB-approved re-consent or written notification. Where an unanticipated problem changes the risk profile of the combined intervention, including a relevant Physical AI safety event, the consent document is updated and currently enrolled participants are re-consented before any further study procedure, consistent with the Physical AI opt-out in the original consent.

10 Statistical Considerations

The statistical framework for this study is that of a confirmatory, randomized, controlled superiority trial designed to test a comparative efficacy hypothesis between the investigational regimen (Arm A) and institutional-standard care (Arm B). Accordingly, the primary analysis is hypothesis-testing and is protected against type I error inflation, in contrast to the predominantly descriptive framework of the single-arm predicate [1]. All analyses follow the principles of the International Council for Harmonisation (ICH) E9 guideline on statistical principles for clinical trials, including the estimand framework for defining the target of estimation, are prespecified in this section and in the Statistical Analysis Plan that this section summarizes, and are reported in accordance with the CONSORT statement for parallel-group randomized trials [27] and the TRIPOD+AI statement for the model-assisted advisory component [28].

10.1 Statistical Hypotheses

The study has one primary efficacy hypothesis and a prespecified hierarchy of key secondary hypotheses, each matched to a population-level summary and an analytic method.

The primary efficacy hypothesis concerns progression-free survival (PFS). Let θ denote the true hazard ratio (HR) for PFS comparing Arm A with Arm B. The null and alternative hypotheses are $H_0 : \theta \geq 1$ (no PFS benefit of Arm A relative to Arm B) versus $H_1 : \theta < 1$. The hypothesis is tested by a stratified log-rank test using the randomization strata (resectability, KRAS allele, neoadjuvant therapy, and site), and the treatment effect is estimated by a stratified Cox proportional-hazards model that yields the HR with a two-sided 95% confidence interval. The test is two-sided at the $\alpha = 0.05$ level, with the type I error allocated across one interim and the final analysis by a group-sequential alpha spending function (§9.2, §9.4.6).

The key secondary hypotheses are tested only if the primary PFS hypothesis is rejected, in a fixed-sequence (hierarchical) order that controls the family-wise type I error at two-sided 0.05 without further multiplicity adjustment: overall survival (OS); the margin-negative (R0) resection rate; the clinically relevant (ISGPS Grade B or C) postoperative pancreatic fistula rate [11]; the major pathologic response (MPR) rate; and the week-12 KRAS ctDNA clearance rate [12]. Each is tested at two-sided 0.05, and testing proceeds down the sequence only while each successive null hypothesis is rejected; the first non-rejection stops the confirmatory sequence, and endpoints below that point are reported descriptively. The remaining secondary endpoints are estimation targets reported with two-sided 95% confidence intervals and are not part of the confirmatory hierarchy. ICH E9 estimand attributes (population, endpoint, handling of intercurrent events such as open conversion, withdrawal, new anticancer therapy, or death, and the population-level summary) are specified for every endpoint in the Statistical Analysis Plan.

10.2 Sample Size Determination

The planned sample size is $n = 220$ participants randomized 1:1 (110 per arm). The sample size is event-driven: under proportional hazards, approximately 140 PFS events provide 85% power at a two-sided $\alpha = 0.05$ to reject the primary null hypothesis when the true PFS hazard ratio is 0.60, corresponding to a median PFS of 14.0 months in Arm B and 23.3 months in Arm A. The randomization of 220 participants is projected to accrue the required 140 events over approximately a 30-month accrual period across the eight sites followed by a minimum of 18 months of additional follow-up. To preserve the evaluable sample against non-evaluability (major eligibility violations, withdrawal before any post-baseline assessment, or missing the primary assessment), up to approximately 245 participants may be enrolled, allowing for up to 10% non-evaluability while retaining 220 evaluable randomized participants.

A single interim analysis for efficacy and futility is performed when approximately 60% of the targeted PFS events (approximately 84 events) have accrued. The type I error is controlled across the interim and final analyses by the Lan-DeMets alpha spending function approximating the O'Brien-Fleming boundary, which spends little alpha at the interim and preserves close to the full two-sided 0.05 for the final analysis [13, 26]; a non-binding futility boundary based on the corresponding beta spending is evaluated at the same interim. The operating characteristics of the design are summarized in Table 7.

Table 7: Design and operating characteristics for the confirmatory progression-free-survival comparison. The trial is an event-driven, randomized, two-arm superiority design with one group-sequential interim analysis governed by a Lan-DeMets O'Brien-Fleming alpha spending function.

Parameter	Planned value	Basis and operating characteristic
Design	Randomized 1:1, two-arm, superiority	Multicenter (8 sites), controlled, open-label with blinded independent central review.
Arms	Arm A vs Arm B	Arm A: daraxonrasib RP2D plus LLM-directed robotic Whipple. Arm B: modified FOLFIRINOX plus standard pancreaticoduodenectomy [15].
Allocation	1:1 (110 per arm)	Central permuted-block randomization, stratified by resectability, KRAS allele, neoadjuvant therapy, and site.
Primary endpoint	Progression-free survival	Time from randomization to RECIST 1.1 progression or death; stratified log-rank with stratified Cox HR.
Target hazard ratio	HR $= 0.60$	Alternative hypothesis ($H_1 : \theta < 1$); corresponds to the assumed treatment benefit.
Median PFS	14.0 vs 23.3 months	Arm B (control) versus Arm A (experimental) under proportional hazards.
Power	85%	At HR $= 0.60$ and two-sided $\alpha = 0.05$.
Significance level	$\alpha = 0.05$ (two-sided)	Allocated across one interim and the final analysis by alpha spending.
Target events	approximately 140 PFS events	Event-driven primary analysis; database lock at the target event count.
Interim analysis	one, at approximately 60% of events (~ 84)	Efficacy and futility; Lan-DeMets O'Brien-Fleming spending; DSMB review [13, 26].
Sample size	$n = 220$ (up to ~ 245 enrolled)	Up to ~ 245 enrolled to allow $\leq 10\%$ non-evaluability.
Accrual and follow-up	~ 30-month accrual, ≥ 18-month follow-up	Across 8 sites; follow-up to the 24-month overall-survival endpoint.

10.3 Populations for Analyses

Four analysis populations are defined and applied consistently across all analyses. The intention-to-treat (ITT) population comprises all randomized participants analyzed according to the arm to which they were randomized, and is the primary population for the primary and key secondary efficacy analyses. The modified intention-to-treat (mITT) population comprises all randomized participants who received any study intervention and contributed at least one post-baseline assessment, and supports sensitivity analyses. The per-protocol population comprises participants who received the assigned intervention without a major protocol deviation that could affect the efficacy assessment, and supports supportive analyses of the primary and surgical-quality endpoints. The safety population comprises all participants who received any study intervention, analyzed according to the intervention actually received, and is the population for all safety analyses. The derivation of the populations and the group-sequential interim with the DSMB efficacy and futility boundaries are shown in Figure 18.

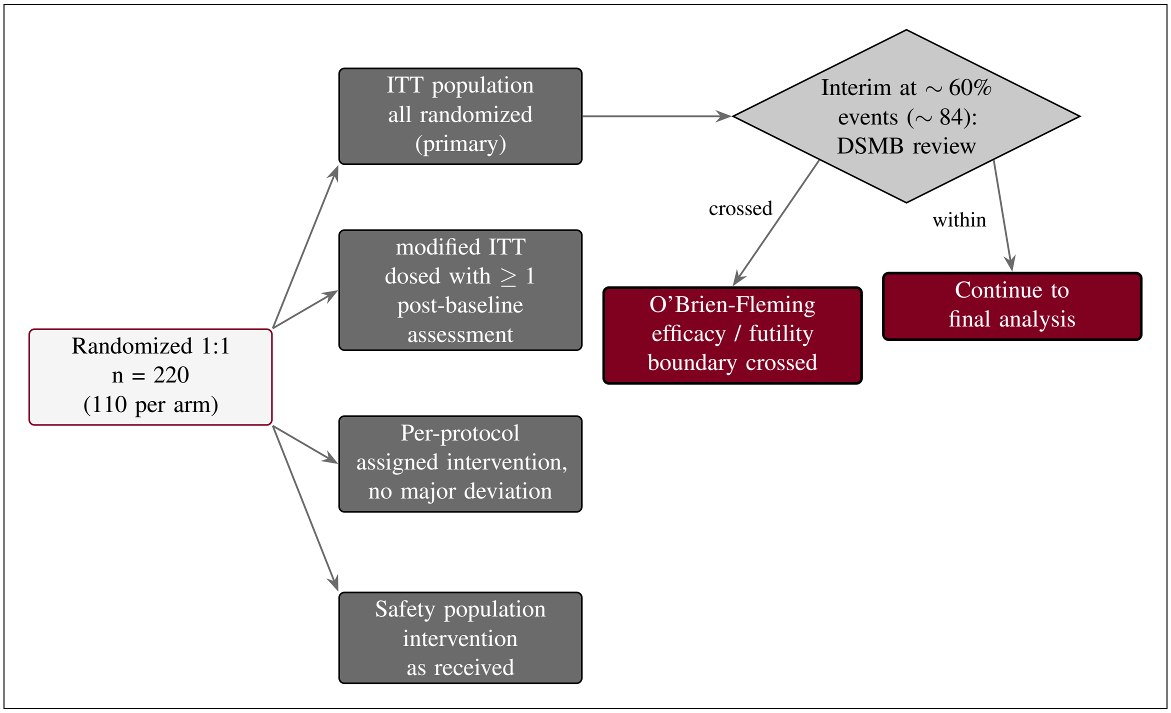

Figure 18. The four analysis populations (intention-to-treat, modified intention-to-treat, per-protocol, safety) and the single group-sequential interim analysis at approximately 60% of the targeted progression-free-survival events, at which the DSMB evaluates the Lan-DeMets O'Brien-Fleming efficacy and futility boundaries. Binding device-safety halt rules operate in parallel: a device-related serious-adverse-event rate breach or a hard-limit breach triggers a mandatory DSMB review.

10.4 Statistical Analyses

10.4.1 General Approach

The primary and key secondary efficacy analyses are confirmatory and are protected against type I error inflation; the remaining analyses are estimation-based and descriptive. Time-to-event endpoints are analyzed by the stratified log-rank test and summarized by the Kaplan-Meier method with the number at risk, and the treatment effect is estimated by a stratified Cox proportional-hazards model reporting the hazard ratio with a two-sided 95% confidence interval; the proportional-hazards assumption is assessed and, if it is not supported, a prespecified alternative (restricted mean survival time) is reported. Binary endpoints are summarized as proportions with two-sided 95% confidence intervals and compared by a stratified test (Cochran-Mantel-Haenszel) where a confirmatory comparison is specified. Continuous variables are summarized by the number of non-missing observations, mean, standard deviation, median, minimum, and maximum; categorical variables are summarized by counts and percentages. ICH E9 estimands govern the handling of intercurrent events. Missing data are handled by the prespecified primary estimand strategy, with sensitivity analyses to assess robustness. All analyses are performed in a validated, version-controlled statistical environment under a deterministic random seed, and the analysis code and derived data sets are deposited so that every reported result is reproducible.

10.4.2 Analysis of the Primary Endpoint

The primary endpoint, PFS, is analyzed on the ITT population. The primary analysis is the stratified log-rank test of $H_0 : \theta \geq 1$ against $H_1 : \theta < 1$ at the alpha allocated to the final analysis by the group-sequential spending function, with the hazard ratio and its two-sided 95% confidence interval estimated from the stratified Cox model. The primary read is the investigator-assessed PFS; a prespecified sensitivity analysis repeats the test and estimation on the BICR-assessed PFS to confirm that the open-label delivery does not propagate into endpoint ascertainment (§8.1). Additional sensitivity analyses repeat the primary analysis on the per-protocol and mITT populations and under alternative censoring rules for new anticancer therapy.

10.4.3 Analysis of the Key Secondary Endpoints

The key secondary endpoints are analyzed only if the primary null hypothesis is rejected, in the fixed-sequence hierarchy of §9.1: OS, then the R0 resection rate, then the ISGPS Grade B or C fistula rate, then the MPR rate, then the week-12 KRAS ctDNA clearance rate. OS is analyzed by the stratified log-rank test and the stratified Cox model on the ITT population; the binary key secondary endpoints are analyzed by the stratified Cochran-Mantel-Haenszel test with the risk difference and its 95% confidence interval. Each test is conducted at two-sided 0.05, and the sequence stops at the first non-rejection. The key secondary and the remaining secondary endpoints, their analysis populations and methods, and the external Dutch 2025 nationwide robotic-pancreatoduodenectomy descriptive context are summarized in Table 8 [10].

Table 8: Key secondary (confirmatory, fixed-sequence) and other secondary (estimation) endpoints, the analysis population and method, and the reporting and external comparator. The Dutch 2025 nationwide robotic-pancreatoduodenectomy cohort is an external descriptive reference, not an inferential comparator.

Endpoint	Population / method	Reporting and comparator
Overall survival (key)	ITT; stratified log-rank, Cox HR	First in the hierarchy after PFS; KM curves, HR with 95% CI; 24-month rate.
R0 resection rate (key)	ITT; stratified CMH	Risk difference with 95% CI; central masked synoptic pathology.
ISGPS Grade B/C fistula (key)	ITT; stratified CMH	Risk difference with 95% CI; graded per the 2016 ISGPS update [11].
Major pathologic response (key)	ITT; stratified CMH	Risk difference with 95% CI; central pathology grading.
Week-12 KRAS ctDNA clearance (key)	ITT; stratified CMH	Risk difference with 95% CI; central laboratory assay [12].
Clavien-Dindo grade III+	Safety; proportion, 95% CI	Estimation only; Dutch 2025 reference 41.3% [10].
90-day mortality	Safety; proportion, 95% CI	Estimation only; Dutch 2025 reference 3.9% [10].
Conversion to open	Safety; proportion, 95% CI	Estimation only; Dutch 2025 reference 10.1% [10].
Operative time, EBL, length of stay	Per-protocol; descriptive	Mean / median with range; Dutch 2025 descriptive context [10].
Quality of life	mITT; descriptive / mixed model	EORTC QLQ-C30 and PAN26 over time [16].
12- and 18-month PFS rate; 24-month OS rate	ITT; Kaplan-Meier	Landmark rates with 95% CI.

10.4.4 Safety Analyses

All safety analyses are performed on the safety population, with events analyzed according to the intervention actually received. Adverse events are coded to the Medical Dictionary for Regulatory Activities (MedDRA) and tabulated by system organ class and preferred term, by maximum severity using the Common Terminology Criteria for Adverse Events (CTCAE), and by relationship to the drug, to the device (Arm A), and to the procedure, with proportions accompanied by 95% confidence intervals and reported by arm. Serious adverse events and deaths are listed individually. Device telemetry constitutes a second, parallel safety stream unique to Arm A. The cyber-physical record is summarized per procedure and in aggregate: emergency-stop activations and the measured stop latency against the ≤ 500 ms requirement; positional and force excursions against the per-arm ≤ 3 N and cumulative ≤ 18 N caps; vascular safety-zone gating events; and the counts of LLM advisories that were accepted, blocked, or escalated to the human operator. A dedicated Physical AI adverse-event tabulation records, for every event, whether an LLM advisory, a robot command, a safety-limit trip, or a human override was implicated, so that events arising from the autonomy-bearing layer are visible and traceable to a deterministic seed and a repository commit through the federated hash-chained audit trail (§8.3.6). Laboratory data, vital signs, and physical-examination findings are summarized descriptively with shift tables relative to baseline.

10.4.5 Baseline Descriptive Statistics

Baseline characteristics are summarized descriptively by arm and overall. Reported variables include demographics (age, sex, race, ethnicity); disease staging and resectability category (resectable versus borderline-resectable); the specific KRAS G12 mutation; baseline CA 19-9; Eastern Cooperative Oncology Group (ECOG) performance status; prior neoadjuvant therapy and number of cycles; and relevant comorbidity and organ-function indices. Consistent with CONSORT and the randomized design, baseline characteristics are presented by arm without a formal significance test of baseline balance [27]; any notable imbalance in a strong prognostic factor is addressed by a prespecified covariate-adjusted sensitivity analysis of the primary endpoint.

10.4.6 Planned Interim Analyses

A single formal interim analysis of the primary endpoint is conducted when approximately 60% of the targeted PFS events (approximately 84 events) have accrued, and is reviewed by the independent DSMB. Efficacy is assessed against the Lan-DeMets O'Brien-Fleming efficacy boundary and futility against the corresponding non-binding futility boundary, so that the cumulative two-sided type I error across the interim and final analyses is controlled at 0.05 [13, 26]. In parallel with the efficacy interim, binding device-safety halt rules operate throughout the trial: a device-related serious adverse-event rate that reaches the prespecified halt threshold, or a breach of a hard cyber-physical safety limit (emergency-stop latency, or the per-arm or cumulative force caps), triggers a mandatory DSMB review and possible halt, and a halt for an immediate hazard is implemented at once under §312.30(h). No efficacy stopping decision is taken outside the prespecified group-sequential boundary.

10.4.7 Sub-Group Analyses

Prespecified subgroup analyses of the primary endpoint are conducted by resectability category, KRAS allele (G12D versus other G12), neoadjuvant therapy (yes versus no), and site, with the hazard ratio and 95% confidence interval displayed in a forest plot and a test of treatment-by-subgroup interaction reported. These analyses are prespecified but are interpreted with caution as supportive and hypothesis-refining; they are not adjusted for multiplicity, and no confirmatory conclusion is drawn from any single subgroup.

10.4.8 Tabulation of Individual Participant Data

Individual participant data are tabulated to support the aggregate analyses. Participant-level listings are provided for demographics and baseline characteristics; randomized arm, stratification factors, and actual intervention received; all adverse events with onset, severity, seriousness, relationship, and outcome; serious adverse events and deaths; the efficacy endpoints and their event or assessment dates; and, for Arm A, the per-procedure device-telemetry and Physical AI event summaries, each keyed to participant and timepoint. Listings are presented so that every aggregate statistic can be traced to its contributing records.

10.4.9 Exploratory Analyses

Exploratory analyses are non-confirmatory and are clearly labeled as such. They include the concordance between LLM advisories and the verdicts of the deterministic safety gates; the sim-to-real gap between the Phase 0 simulation predictions and the observed intraoperative telemetry; the performance of the federated-learning model across sites over the course of the trial; the longitudinal dynamics of KRAS ctDNA beyond the week-12 clearance endpoint [12]; and a health-economic and cost-effectiveness analysis linked to the co-investment facility. These exploratory results generate hypotheses for later development and are not used to draw confirmatory efficacy conclusions in this protocol.

11 Regulatory, Ethical, and Oversight Considerations

The oversight architecture for this multicenter randomized controlled study layers a conventional human-subject-protection structure with the additional governance that an autonomy-bearing surgical system, deployed as a harmonized fleet across eight sites, requires. A central Coordinating Center directs the conduct of the trial across the network, and the Sponsor-Investigator, ChemicalQDevice, answers to a single Institutional Review Board (sIRB) of record under 45 CFR 46.114 [7] and to an independent data and safety monitoring board (DSMB) with a group-sequential interim plan. The structure convenes a Physical AI Safety Review Committee on a $\leq$ 90-day cadence, designates an Independent Safety Monitor (ISM) for each procedure, and delegates day-to-day monitoring to a robotics- and artificial-intelligence-competent clinical research organization (CRO) over the site Principal Investigators and their teams. Figure 19 presents the multicenter governance and safety-oversight structure.

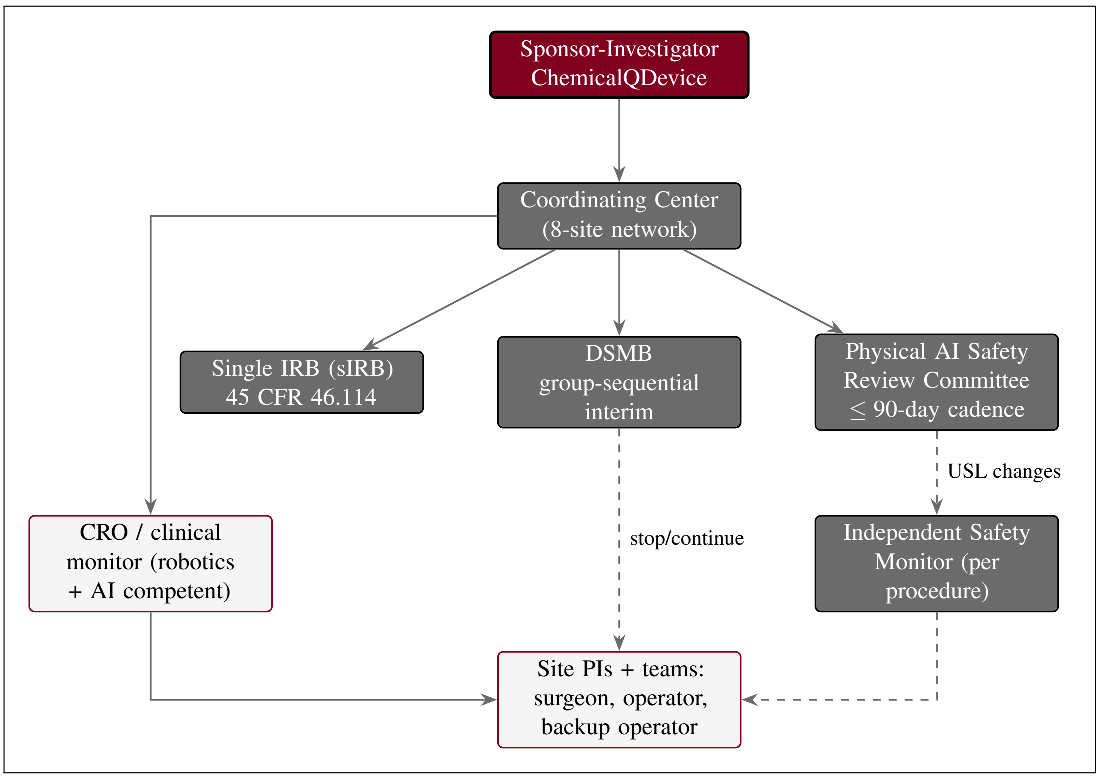

Figure 19. Multicenter governance and safety oversight. The Sponsor-Investigator directs the trial through a Coordinating Center across the eight-site network and answers to a single IRB and an independent DSMB with a group-sequential interim plan, convenes a Physical AI Safety Review Committee on a $\leq$ 90-day cadence, designates an Independent Safety Monitor for each procedure, and delegates monitoring to a robotics- and AI-competent CRO over the site Principal Investigators and their teams.

11.1 Informed Consent Process

Informed consent is obtained by a qualified member of the research team, who is not the treating surgeon, before any study-specific procedure, in language the participant can understand and with adequate time and opportunity to ask questions, consistent with 21 CFR part 50 [2] and ICH E6(R3). A single master consent form is used across the network under the single IRB. The consent discussion covers the diagnosis, the alternatives, and the prognosis; the randomized 1:1 allocation between Arm A (daraxonrasib at the recommended Phase 2 dose plus the LLM-directed robotic Whipple) and Arm B (modified FOLFIRINOX plus standard high-volume pancreaticoduodenectomy) and the open-label nature of that allocation; the risks of each regimen; and, in addition to the standard elements, an explicit and distinct disclosure of the on-premises LLM-directed robotic system, its role and continuous human oversight, and its system-specific risks, as required for an autonomy-bearing investigational device. The consent form addresses the therapeutic misconception directly and explains the staged-autonomy model under which the system operates only at the supervised semiautonomous level.

A defining feature of this trial class is the documented Physical AI opt-out. Under 21 CFR §312.60(f), as adapted for Physical AI investigations [6], a participant randomized to Arm A is offered, and may exercise, a documented right to request administration without the Physical AI advisory layer where clinically feasible, without penalty and without affecting eligibility for the remainder of the study or for clinical care. The participant's election or declination of the opt-out is recorded in the source document and the CRF. A legally authorized representative process and the use of qualified interpreters and translated materials for participants with limited English proficiency are provided so that consent is meaningful for every eligible individual across the network. The right to withdraw at any time without penalty is stated at consent and reaffirmed at each contact. Figure 20 shows the consent process and the opt-out.

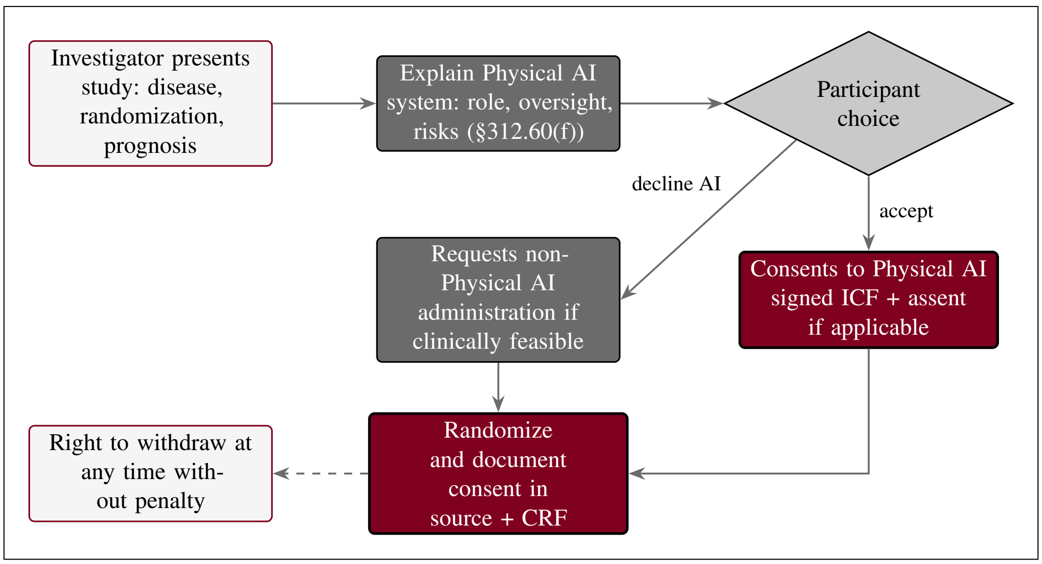

Figure 20. Informed-consent process with the Physical AI opt-out. Consent discloses the randomized allocation and the Physical AI system's role, continuous human oversight, and system-specific risks, and offers a participant randomized to Arm A a documented right to request non-Physical AI administration where clinically feasible (21 CFR §312.60(f)), with the standard right to withdraw.

11.2 Funding, Co-Investment Governance, and the Capital Firewall

This Phase 2 trial is funded through a Patient-Aligned Co-Investment Facility (PACIF), a mechanism by which wealthier individuals connected to cancer patients may contribute ring-fenced philanthropic and private capital with the sole and explicit purpose of raising the likelihood of trial success through operational capacity. The facility purchases operational capacity only, and never scientific outcomes: capital is directed exclusively to six operational levers, namely (1) the eight-site academic HPB network, (2) the expanded Phase 0 simulation compute required to meet the upgraded readiness gate, (3) the harmonized robot fleet across sites, (4) the Patient Access and Equity Fund, (5) the central review and biomarker core, and (6) the independent oversight bodies. None of these levers touches the science of the comparison.

The integrity of the comparison is protected by a capital firewall, a structural separation that prevents any funder or contributor from influencing the randomization, the endpoint definitions, the outcome adjudication, the statistical analysis, the data access, or the publication of the trial. Contributions enter through an independent fund administrator and are disbursed only against the permitted operational levers; no funder holds any role, vote, or veto over the scientific conduct, and no contribution is conditioned on any result. All financial interests are disclosed and managed under 21 CFR part 54 [3], and the facility is governed by the verification-and-validation-and-uncertainty-quantification (VVUQ) financial-data standard of the H.R. 9510 framework [8], which ties the funding and oversight of Physical AI oncology investigations to a verifiable evidentiary standard. The Patient Access and Equity Fund is administered for equitable disbursement, supporting participation costs and access across the network, and is itself walled off from eligibility determinations and outcome ascertainment, so that no participant's access to the trial or to its assessments can be influenced by the source of funds. All results, regardless of outcome, are openly deposited under the Creative Commons Attribution 4.0 International (CC BY 4.0) license. Figure 21 shows the capital firewall governance.

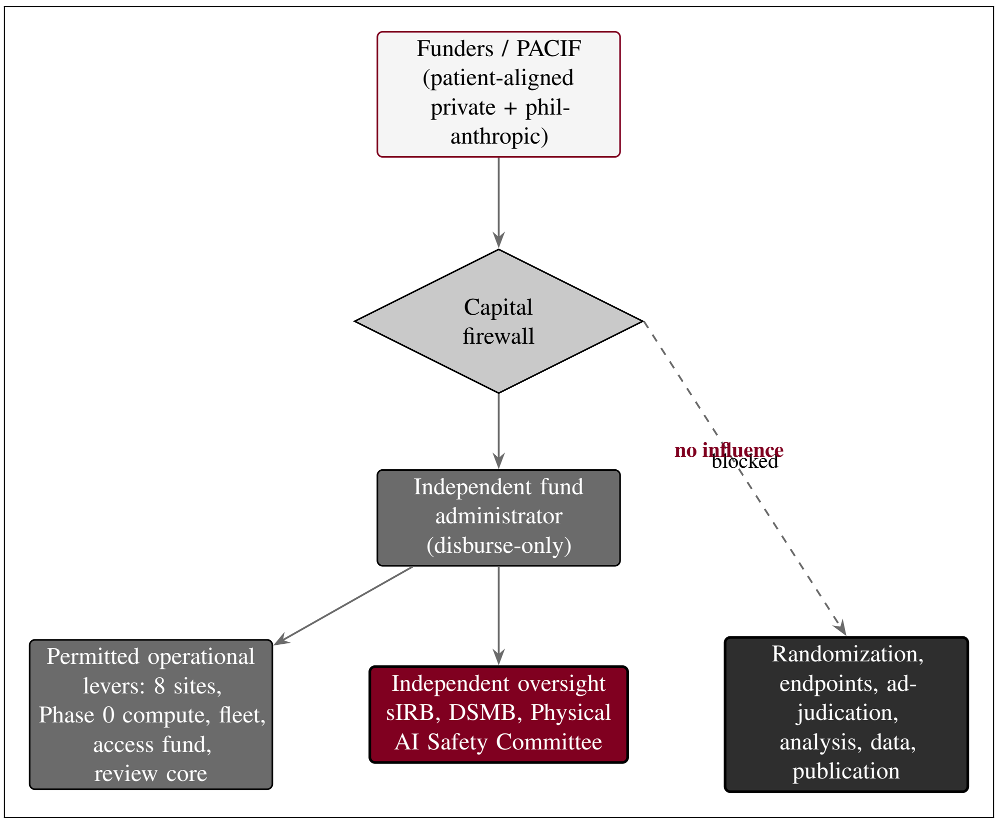

Figure 21. Capital firewall governance. Patient-aligned funders contribute through the PACIF into an independent fund administrator behind a capital firewall that disburses only to the permitted operational levers (the eight-site network, Phase 0 compute, the robot fleet, the Patient Access and Equity Fund, and the central review and biomarker core) and to independent oversight; the dashed blocked path shows that funders cannot reach randomization, endpoint definition, outcome adjudication, statistical analysis, data access, or publication.

11.3 Study Discontinuation and Closure

The study may be discontinued or closed by the Sponsor-Investigator, by the single IRB, or by the FDA. Grounds for closure include unacceptable safety findings (including the prespecified DSMB stopping rules of §9), a clinical hold, a determination that the study cannot be conducted in compliance, or completion of the planned accrual and analysis. The FDA may impose a clinical hold on the drug or device component under the provisions for terminating or suspending an investigation; in addition, the Physical AI overlay establishes specific clinical-hold grounds for the autonomy-bearing system, including a breach of a hard cyber-physical safety limit, a failure of the verification-before-generation gate surface, or a loss of audit integrity, and these grounds may halt activity at a single site or across the fleet. On any hold or closure, enrollment and patient-facing robotic activity stop immediately at the affected scope; affected participants are transitioned to appropriate clinical care; and a controlled decommissioning and data-preservation procedure is executed, under which the robotic system is locked out of patient contact, the deterministic configuration and repository commit are recorded, and the complete federated hash-chained command and telemetry record is sealed and retained for the regulatory retention period (§10.10). The single IRB, the FDA, and the DSMB are notified, and a final report is prepared.

11.4 Confidentiality and Privacy

Participant confidentiality is protected throughout. Data released for analysis, deposition, or publication are de-identified by the HIPAA Safe Harbor method, with removal of all eighteen categories of protected health information identifiers (names; geographic subdivisions smaller than a state; all date elements other than year that are directly related to an individual, and ages over 89; telephone and fax numbers; email addresses; Social Security numbers; medical record numbers; health plan beneficiary numbers; account numbers; certificate or license numbers; vehicle identifiers; device identifiers and serial numbers; web URLs; IP addresses; biometric identifiers; full-face photographs and comparable images; and any other unique identifying number, characteristic, or code). Access to identifiable records is governed by a deny-by-default authorization model, in which no role can read, write, or export a record except where an access right has been explicitly granted, and every access, change, and export is written to a federated hash-chained audit trail in which each entry cryptographically incorporates the hash of the prior entry so that the record is tamper-evident and any alteration is detectable across the eight sites. All electronic records, signatures, and the cyber-physical telemetry stream are maintained under 21 CFR part 11 [5], with validated systems, secure time-stamped audit trails that do not obscure prior entries, role-based access controls, and electronic signatures bound to the records they authenticate.

11.5 Future Use of Stored Specimens and Data

The cyber-physical record is retained as a tiered telemetry pyramid that balances long-term scientific reuse against storage cost. At the apex are the compact, human-readable summaries and the safety-event and audit logs retained in full and indefinitely; the middle tiers hold compressed feature and event streams; and the base, level zero (L0), holds the raw, losslessly reconstructable command and sensor stream. The L0 raw tier is deposited on Zenodo under controlled terms and is decompressible and reconstructable on FDA request as required for inspection and review under 21 CFR §312.57, so that any reported behavior can be replayed deterministically from the seed and the repository commit. Circulating tumor DNA (ctDNA) and other biomarker specimens collected for the central biomarker core are governed by the participant's consent and applicable regulation, are processed for the protocol-specified assessments, and are not retained for an unspecified future purpose without appropriate consent. Any future secondary use of stored de-identified data is limited to research consistent with the consent, is conducted on de-identified data only, and preserves the audit chain.

11.6 Key Roles and Study Governance

The Sponsor-Investigator (ChemicalQDevice, Kevin Kawchak) holds the combined IND/IDE and bears the sponsor and investigator responsibilities under 21 CFR parts 312 and 812, including protocol custody, safety reporting, oversight of the investigation, and assurance of compliance across the network. The Coordinating Center provides central operational direction, the single IRB of record, the central randomization and review services, and harmonization across sites. Each site Principal Investigator directs the conduct of the study at that site and supervises the research team. At each procedure in Arm A, the operating surgeon performs the pancreaticoduodenectomy under the supervised semiautonomy model; a primary operator drives the system, and a qualified backup operator is present and ready to assume control or to invoke the manual override at any time. The CRO provides robotics- and AI-competent clinical monitoring across sites. The DSMB, the Independent Safety Monitor, and the Physical AI Safety Review Committee provide the independent oversight described in §10.9. Roles, delegated duties, and the corresponding qualifications are documented on a delegation-of-authority log maintained at each site, and changes are recorded and reported as required.

11.7 Safety Oversight

Independent safety oversight operates on three tiers. The DSMB, composed of members independent of the conduct of the study with surgical, oncologic, biostatistical, and device-safety expertise, reviews accumulating safety and efficacy data, executes the prespecified group-sequential interim analysis with O'Brien-Fleming spending [13], applies the prespecified stopping rules of §9, and recommends continuation, modification, pause, or halt for the trial or for an individual site. The Independent Safety Monitor is designated for each procedure and provides real-time, case-level safety oversight independent of the operating team, with the authority to call a stop. The Physical AI Safety Review Committee meets on a $\leq$ 90-day cadence (and ad hoc on any triggering event) to review the autonomy-bearing behavior of the fleet: the telemetry and audit record, the verification-before-generation gate-surface decisions, the Unified Safety Level (USL) rating, and any software change, and to recommend USL reassessment or oversight changes to the Sponsor-Investigator. The three tiers are coordinated so that no autonomy-bearing change and no escalation decision proceeds without appropriate independent review.

11.8 Clinical Monitoring

The study is monitored by a CRO whose monitors are competent in both clinical-trial monitoring and the robotics and artificial-intelligence elements of this trial class. Monitoring follows a risk-based plan calibrated to site volume and risk, and includes source-data verification not only of conventional clinical source documents and the consent records but also of the cyber-physical evidence at each site: the device telemetry, the verification-before-generation gate-surface logs, and the federated hash-chained audit trail, which are verified against the recorded outcomes and the deterministic seed and repository commit. Monitoring visits verify eligibility, randomization integrity, consent (including the documented Physical AI opt-out election in Arm A), safety reporting, protocol adherence, and data integrity across the network, and findings are documented and resolved through to closure.

11.9 Quality Assurance and Quality Control

Quality assurance and quality control span the clinical and the cyber-physical domains and are harmonized across the eight sites. Before every procedure in Arm A, the pre-procedure safety matrix is verified, the USL readiness rating is confirmed at or above the deployment threshold, and the emergency-stop, positional, and force envelopes are checked. After any software update, model change, or configuration change to the LLM-directed system, a Phase 0 re-validation is performed against the independent simulation frameworks, the USL is reassessed against the upgraded threshold of at least 8.0, and fleet harmonization is reconfirmed so that every device instance behaves identically, before any further patient-facing use at any site; no update reaches a participant without re-validation. Audits may be conducted by the Sponsor-Investigator or a designee independent of the conduct of the study, and corrective and preventive actions are documented. The deterministic, version-controlled build ensures that the validated configuration in use is exactly the configuration that was tested and recorded.

11.10 Data Handling and Record Keeping

Seven Physical AI record types are captured and retained for this trial class, in addition to the conventional clinical trial records: (1) the LLM advisory record (each prompt, context, and advisory output); (2) the robot command record (each command issued to the arms); (3) the sensor and telemetry record (the cyber-physical stream, tiered as in §10.5); (4) the safety-limit-event record (emergency-stop activations and latencies, positional and force-envelope excursions, and vascular-exclusion gating events); (5) the verification-before-generation gate-surface record (each accept, block, or escalate decision); (6) the human-override and intervention record; and (7) the federated hash-chained audit trail that binds the preceding six to a deterministic seed and a repository commit across the network. All records are retained for the period required by 21 CFR §312.57 (at least two years after a marketing application is approved for the indication, or, if no application is filed or approved, two years after the investigation is discontinued and the FDA is notified), and the L0 raw tier is preserved as described in §10.5 so that a requested record can be reconstructed.

11.11 Protocol Deviations

A protocol deviation is any departure from the IRB-approved protocol. Deviations are identified, documented, assessed for their effect on participant safety and data integrity, and reported to the single IRB and the Sponsor-Investigator in accordance with the IRB's policy and applicable regulation; deviations that affect safety or the rights and welfare of participants, and any that recur, are escalated, and the Coordinating Center tracks deviation patterns across sites. Deviations are distinguished from planned protocol amendments, which follow the amendment process. Where a deviation is necessary to eliminate an apparent immediate hazard to a participant, the change may be implemented without prior IRB or FDA approval under 21 CFR §312.30(b) (and the corresponding device provision), with prompt notification and reporting afterward; for the autonomy-bearing system, an immediate-hazard modification also triggers the decommissioning and re-validation controls of §10.4 and §10.11 before any further patient-facing use.

11.12 Publication and Data Sharing Policy

Results are disseminated regardless of outcome through dual deposition: the analysis code, derived data sets, and protocol are published in a public GitHub repository and archived on Zenodo with a citable DOI, under the CC BY 4.0 license. Reporting follows the TRIPOD+AI statement for the model-assisted advisory component [28] and the CONSORT statement for the parallel-group randomized design [27], in addition to the applicable clinical-trial reporting standards. The deterministic seed and the repository commit are published so that every reported result is reproducible, and the tiered telemetry record is deposited as described in §10.5. The results of the co-investment-funded trial are openly deposited so that the funding model is transparent and the science is independently verifiable. No identifiable participant information is published; all shared data are de-identified by the HIPAA Safe Harbor method (§10.5).

11.13 Conflict of Interest Policy

Financial interests are disclosed and managed in accordance with 21 CFR part 54 (Financial Disclosure by Clinical Investigators) [3]. The Sponsor-Investigator is the chief executive of ChemicalQDevice, which develops the Physical AI system under study; this relationship is disclosed in full, and the independent oversight structure of §10.9 (the DSMB, the Independent Safety Monitor, and the Physical AI Safety Review Committee) is established in part to manage it, so that safety and escalation decisions are not made by a party with a financial interest in the outcome alone. The co-investment capital is managed behind the capital firewall described in §10.3, so that no funder, including any with a financial interest, can influence the randomization, the endpoints, the adjudication, the analysis, the data access, or the publication. Investigator financial-disclosure certifications are collected before participation and updated through one year after the study, and any identified conflict is disclosed and managed before it can affect the conduct, analysis, or reporting of the study.

12 Additional Considerations, Abbreviations, and Amendment History

12.1 Additional Considerations

Although this protocol is written to the United States combined IND/IDE pathway, the LLM-directed robotic system is a Software as a Medical Device (SaMD) whose deployment may be sought across jurisdictions, and the design anticipates the parallel regulatory tracks summarized in Table 9. The European Union Medical Device Regulation (EU MDR) and its companion artificial-intelligence requirements, China's National Medical Products Administration (NMPA), Japan's Pharmaceuticals and Medical Devices Agency (PMDA), the United Kingdom's Medicines and Healthcare products Regulatory Agency (MHRA), Health Canada, and Australia's Therapeutic Goods Administration (TGA) each maintain a SaMD and robotically assisted surgery framework with which the device evidence is designed to remain compatible. The common technical spine, Phase 0 simulation validation, USL readiness rating, the federated hash-chained audit trail, and the verification-before-generation gate surface, is constructed to map onto each track without redesign, and the international consensus standards cited in §12 (IEC 80601-2-77, ASME V and V 40, NASA-STD-7009A, IEEE 7009-2024) provide a shared basis across jurisdictions [29, 30, 31, 32].

A multicenter Physical AI trial also depends on learning across sites without centralizing raw patient data. The eight academic HPB centers operate as a federated network: model and validation evidence is exchanged across sites while raw telemetry and identifiable records remain local, the audit trail is hash-chained across the fleet so that every site's record is tamper-evident, and fleet harmonization ensures that every device instance behaves identically. This federated design lets the network accumulate validation evidence and reproduce simulated procedures at scale while preserving the deny-by-default privacy model of §10.6 and the upgraded Phase 0 readiness gate at each site.

The trial is additionally aligned with the legislative and financial framework of the Verification Before Generation in Physical AI Oncology Trials Act of 2026 (H.R. 9510) [8], which ties the funding and oversight of Physical AI oncology investigations to a verification-and-validation-and-uncertainty-quantification (VVUQ) evidentiary standard. That framework supplies the financial-data standard under which the Patient-Aligned Co-Investment Facility operates: co-investment capital raises operational capacity and trial success likelihood while the capital firewall and the VVUQ standard together ensure that funding is verifiable and that no funder can reach the science of the comparison. The ten-gate assurance suite operationalizes the technical half of that standard: every proposed robot-patient command is verified before it is generated, against fourteen external safety and robotics standards plus two clinical baselines, with epistemic and aleatory uncertainty bounds, and the gate surface returns one of accept, block, or escalate. Figure 22 renders the ten-gate pipeline.

Table 9: Cross-jurisdictional Software as a Medical Device (SaMD) and robotically assisted surgery tracks. The common technical spine (Phase 0 validation, USL readiness, the federated hash-chained audit trail, and the verification-before-generation gate surface) maps onto each track without redesign.

Jurisdiction	Regulator / framework	Alignment basis
European Union	EU MDR + AI reqs	SaMD risk classification; common simulation-validation and federated audit evidence.
China	NMPA	SaMD, robotically assisted surgery review; standards-based VVUQ.
Japan	PMDA	SaMD and device review; consensus-standard alignment.
United Kingdom	MHRA	SaMD framework; post-market and audit-trail compatibility.
Canada	Health Canada	SaMD licensing; quality-system and validation evidence.
Australia	TGA	SaMD inclusion; standards-based safety and performance evidence.

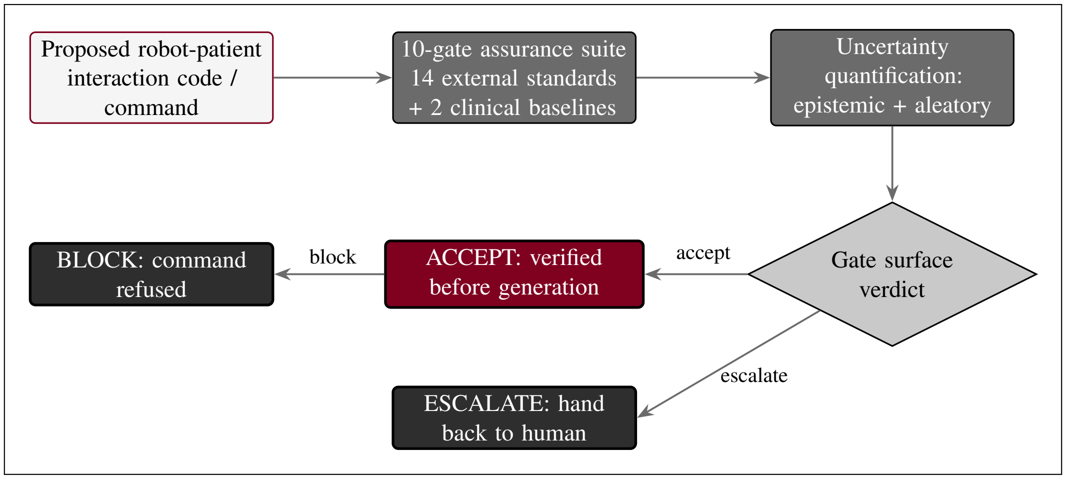

Figure 22. Verification before generation. Every proposed robot-patient command passes a ten-gate assurance suite aligned to fourteen external safety and robotics standards plus two clinical baselines, with epistemic and aleatory uncertainty bounds, before the gate surface returns ACCEPT, BLOCK, or ESCALATE (hand back to human).

12.2 Protocol Amendment History

Table 10 records the version history of this protocol and its Phase 1 predicate; subsequent amendments will be appended with the version, date, sections affected, and a summary of the change.

Table 10: Protocol amendment history.

Version	Date	Sections affected	Summary of change
v1.0.0	June 20, 2026	All (predicate protocol)	Phase 1 predicate protocol [1]: the first-in-human, single-arm, combined IND/IDE design that established the daraxonrasib recommended Phase 2 dose through a perioperative 3+3, the on-premises LLM-directed eight-arm robotic Whipple device feasibility, the Physical AI Subpart J overlay, and the safety and oversight framework on which this Phase 2 builds.
v1.1.0	June 23, 2026	All (this protocol)	This Phase 2, multicenter, randomized, controlled protocol. Establishes the randomized 1:1 parallel-group design with BICR, daraxonrasib at the recommended Phase 2 dose plus standard-of-care control, the upgraded Phase 0 readiness gate (USL $\geq$ 8.0), multicenter single-IRB governance, the co-investment model and the capital firewall, and the confirmatory group-sequential statistical framework.

12.3 Abbreviations

Table 11 resolves every abbreviation used across this protocol.

Table 11: Abbreviations used in this protocol.

Abbr.	Definition	Abbr.	Definition
AE	Adverse event	MTD	Maximum tolerated dose
BICR	Blinded independent central review	NMPA	National Medical Products Administration
CONSORT	Consolidated Standards of Reporting Trials	OS	Overall survival
CRF	Case report form	PACIF	Patient-Aligned Co-Investment Facility
CRO	Clinical research organization	PDAC	Pancreatic ductal adenocarcinoma
ctDNA	Circulating tumor DNA	PFS	Progression-free survival
CTCAE	Common Terminology Criteria for Adverse Events	PJ	Pancreaticojejunostomy
DLT	Dose-limiting toxicity	PMDA	Pharmaceuticals and Medical Devices Agency
DSMB	Data and Safety Monitoring Board	PP	Per-protocol
ECOG	Eastern Cooperative Oncology Group	PV	Portal vein
EORTC	European Organisation for Research and Treatment of Cancer	QoL	Quality of life
FDA	U.S. Food and Drug Administration	R0	Margin-negative resection
GCP	Good Clinical Practice	RCT	Randomized controlled trial
GJ	Gastrojejunostomy	RP2D	Recommended Phase 2 dose
HJ	Hepaticojejunostomy	SAE	Serious adverse event
HPB	Hepatopancreatobiliary	SaMD	Software as a Medical Device
HR	Hazard ratio	sIRB	Single Institutional Review Board
ICH	International Council for Harmonisation	SMA	Superior mesenteric artery
IDE	Investigational Device Exemption	SMV	Superior mesenteric vein
IND	Investigational New Drug	TGA	Therapeutic Goods Administration
ISGPS	International Study Group of Pancreatic Surgery	USL	Unified Safety Level
ISM	Independent Safety Monitor	VVUQ	Verification, validation, and uncertainty quantification
ITT	Intention-to-treat	MDR	Medical Device Regulation
KRAS	Kirsten rat sarcoma viral oncogene homolog	MHRA	Medicines and Healthcare products Regulatory Agency
LLM	Large language model	MPR	Major pathologic response
mITT	Modified intention-to-treat		

13 References and Back Matter

The reference list is rendered below through the `ieeetr` bibliography style from `references.bib`, and includes the five daraxonrasib program entries; the three main documents on which this protocol builds; the Phase 1 predicate protocol; the directly relevant author works; the clinical references; and the FDA guidance, the applicable 21 CFR parts (11, 50, 54, and 812) and 45 CFR 46, and the IEC, ASME, NASA, and IEEE standards with the NIH template.

- The 2030 PDAC sixty-second robotic Whipple and daraxonrasib simulation [18] is the simulation source from which the clinical figures and the operative and telemetry behavior in this protocol derive; it supplies the in silico evidence that the device arm is built upon.
- The Adaption of 21 CFR Part 312 for end-to-end Physical AI oncology trial unification [6] is the regulatory source for the Physical AI overlay, including Subpart J, the §312.60(f) Physical AI consent and opt-out, the Phase 0 simulation-validation gate, the USL readiness rating, and the hash-chained audit-trail requirements adopted throughout.
- The H. R. 9510 bill, Verification Before Generation in Physical AI Oncology Trials Act of 2026 [8], is the legislative and financial source that ties the funding and oversight of Physical AI oncology investigations to the VVUQ evidentiary standard operationalized by the ten-gate assurance suite (§11.1), and supplies the financial-data standard governing the co-investment facility and the capital firewall (§10.3).
- The Phase 1 predicate protocol [1] is the source that established the recommended Phase 2 dose of daraxonrasib and the device feasibility of the on-premises LLM-directed robotic Whipple, on which this randomized, controlled Phase 2 protocol builds.

Acknowledgments

This protocol was assembled with Claude Code (Opus 4.8, 1M context). The author also acknowledges ChatGPT 5.5 (Thinking Extended) and Gemini 3.1 Pro for additional research and editing assistance.

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Ethical Disclosures

This is an independent research draft. No real patients were enrolled, no protected health information was used, and no IRB approval was sought or obtained. All clinical figures derive from the author's simulation sources and the Phase 1 predicate and are illustrative unless tied to a cited reference. This document is not an active IND or IDE, is not an approved protocol, and is not medical or regulatory advice; it is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor.

Data Availability

The protocol, the analysis code, and the derived data sets are available through the public GitHub repository kevinkawchak/physical-ai-oncology-trials/tree/main/trial-phase-2 (v4.1.0) and archived on Zenodo under the Creative Commons Attribution 4.0 International (CC BY 4.0) license https://creativecommons.org/licenses/by/4.0/deed.en. All deposited data are simulation-derived, generated under a deterministic seed, and contain no personally identifiable information; the results of the co-investment-funded trial are deposited openly.

Cite This Article

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